



STANDING UP FOR STILLBIRTH

Current estimates
and key interventions

Report of the United Nations
Inter-agency Group for Child
Mortality Estimation, 2024

This report was prepared at the United Nations Children’s Fund (UNICEF) Headquarters by Lucia Hug, Naomi Lindt, Yang Liu, Wanli Nie, David Sharrow, Graeme Wilson Fell and Danzhen You on behalf of the United Nations Inter-agency Group for Child Mortality Estimation (UN IGME). Danzhen You provided strategic and technical guidance. Special thanks to Gagan Gupta from UNICEF and Kathleen Strong from the World Health Organization (WHO) for providing critical inputs to the report. Thanks also go to the following colleagues for their valuable inputs and comments: Vivian Lopez, Shaimaa Ibrahim and Minjoon Kim from UNICEF; Patrick Gerland and Thomas Spoorenberg from the United Nations Department of Economic and Social Affairs, Population Division; Helena Cruz Castanheira from the United Nations Economic Commission for Latin America and the Caribbean, Population Division; Emi Suzuki from the World Bank Group; and Bochen Cao, Theresa Diaz and Wilson Were from WHO.

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PREVENTING, COUNTING AND HONOURING THE WORLD'S STILLBORN BABIES

Every day, more than 5,000 women around the world endure the heartbreaking experience of stillbirth. Many of these tragedies seem to happen suddenly or without reason – leaving millions of mothers shocked, traumatized and grieving, asking questions that will never be answered.

But the truth is, most stillbirths are preventable when women have access to timely, high-quality care during pregnancy and childbirth. And when a stillbirth does occur, there are many measures health care practitioners can take to support affected women and their families through compassionate care and accurate documentation of the event. These data not only honour the dignity of bereaved parents by providing clarity, closure and the answers they deserve, but they also hold the power to prevent future stillbirths.

More than numbers on a page

From low- to high-income settings, progress has been made in reducing the number of stillbirths that occur each year. This achievement is about far more than numbers on a page – it reflects real children growing, learning and playing today. They are alive because countries have made investments towards ensuring pregnant mothers and their babies can access quality health care.

Children around the world are marking milestones and celebrating birthdays thanks

to global commitments like the World Health Assembly 77 Resolution, which calls on countries to further reduce maternal, newborn and child mortality, and countries' adoption and implementation of initiatives like the Every Woman Every Newborn Everywhere (EWENE)¹ partnership and Child Survival Action.²

These children are living proof that when global health actors champion and finance equitable interventions and national governments commit to adopting, scaling and sustaining them, fewer children will die. Just think: If the world had remained complacent and the stillbirth rate had remained unchanged since 2000, 19 million more babies would have been stillborn – increasing the global tally of deaths over the last two decades by a third. This sobering number underscores the urgent need for sustained and targeted focus, attention and financial investments in stillbirth prevention and care.

Unjust outcomes

But this year's UN IGME stillbirth estimates also sound a clarion call to do far better. For one, they show that progress in reducing the stillbirth rate is slower than it should be – it has not matched the gains made in reducing child mortality. They also highlight that the burden of stillbirth is not equitably distributed. Stillbirth rates vary widely by region, country and within countries; mothers in the poorest settings and fragile- and conflict-affected situations face the highest risks. Factors like

a woman's place of residence, her level of education, household wealth and age also shape her risk of having a stillborn child.

UN IGME's estimates also show that approximately 900,000 babies continue to die each year during labour – a crucial piece of information, considering that most intrapartum stillbirths are preventable.

Furthermore, despite some progress towards understanding women's and families' needs after a stillbirth, national implementation of bereavement care pathways remains limited to a few high-income contexts. The vast majority of the 1.9 million mothers who undergo a stillbirth every year remain deprived of the compassionate care they need to heal from these very real traumas.

A shared commitment, a moral duty

To maintain and advance what the world has achieved in reducing the stillbirth rate, the global community must remain steadfast in its commitment to providing the health funding and measures that reduce and

treat the known causes and risk factors of stillbirths. It must continue to leverage partnerships, know-how and resources to maximize the impact of every dollar spent.

The stillbirth rate acts as a critical indicator of the quality of care available to mothers during pregnancy and childbirth. The ripple effects of stillbirth are long-lasting, carrying emotional, financial and economic consequences that not only affect mothers, but also their families and communities. The global health community has a duty to not only recognize, understand and respond to these deaths – that is, by counting them, properly classifying them and taking action to prevent them – but also to care for the mothers and their families who experience a stillbirth. With strong political will, sound policies and targeted, reliable investments along the continuum of care for every mother and child, the world can turn the tide on stillbirths.

It is time for the global community to definitively address this neglected child rights issue once and for all.



STILLBIRTH: Key findings in this year's report

Stillbirths continue to occur with alarming frequency – even though many of them are preventable

In 2023, **almost 2 million** babies¹ were stillborn during the last trimester of pregnancy (at or after 28 weeks of gestation). An estimated:

14 in every 1,000

babies born in 2023 were stillborn

That's **1 in 70**

babies never taking their first breath



More than **4 in 10** babies **died during labour** (i.e., intrapartum) – a time when many stillbirths can be prevented with the aid of skilled health personnel during childbirth

In the places **where stillbirth rates are highest**, the proportion of intrapartum stillbirths is even higher, estimated at **49 per cent or higher**

Data availability on stillbirths remains inadequate

Every baby deserves to be counted – whether they are born alive or stillborn. Yet stillbirths are frequently excluded from routine data systems, and when these data are included, poor data quality often limits their utility. **One in four countries** does not produce quality data on stillbirths. **Fewer than half of countries** report data in relation to the onset of labour (whether the death occurred during the antepartum or intrapartum period) – vital information that practitioners and policymakers need to target interventions and save lives. Current estimates are limited to late gestation stillbirths (in the third trimester, i.e., 28 weeks or later) due to methodological and data inconsistencies at earlier gestational ages.

These data gaps weaken the accurate monitoring of stillbirth rates around the world and inhibit the formulation of the policies needed to prevent stillbirths. Improving stillbirth reporting is a crucial step towards ensuring that every baby counts. To close the data gaps, countries must:



Strengthen health systems at national and subnational levels to capture stillbirth data from 22 weeks gestation



Standardize definitions and reporting practices to align with international standards, ensuring consistency and comparability across countries

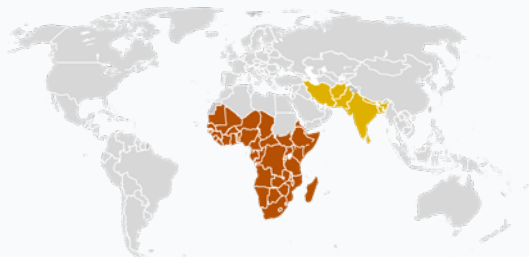


Address stigma and reporting challenges, as fear of blame and inconsistent classification often lead to the underreporting and misclassification of stillbirths

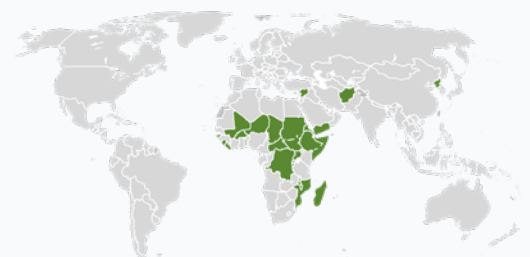
Mothers in low-income countries, sub-Saharan Africa and Southern Asia, and fragile- and conflict-affected situations face the highest risks of experiencing the tragedy of stillbirth

Almost **one in three (32 per cent)** stillbirths in 2023 took place in low-income countries, and nearly **80 per cent** occurred in sub-Saharan Africa and Southern Asia.³ The number and share of stillbirths are on the rise in sub-Saharan Africa, where nearly **48 per cent** of the world's stillbirths took place in 2023, versus only **30 per cent** of global live births: The highest national stillbirth rate in 2023 was **≥ 21 times** that of the lowest national rate.

In fragile and conflict-affected situations (FCS),⁴ the stillbirth rate is **twice as high** as that of more stable regions. While 25 per cent of live births occur in these settings, FCS countries accounted for **41 per cent** of stillbirths in 2023.



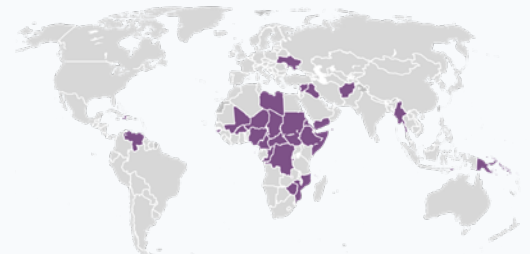
Nearly **8 in 10 (79%)** stillbirths occur in sub-Saharan Africa and Southern Asia



Low-income countries account for **32%** of stillbirths, but only **19%** of live births



Sub-Saharan Africa accounts for **nearly half (48%)** of global stillbirths, but only **30%** of live births



FCS countries: **41%** of stillbirths, **25%** of live births

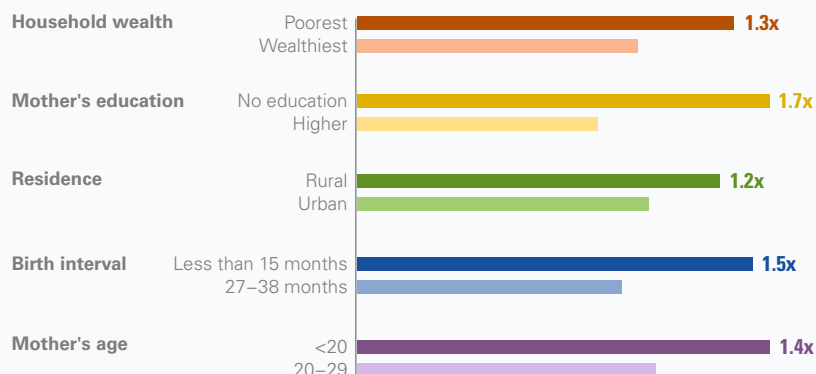
Note: Map does not reflect a position by UN IGME agencies on the legal status of any country or territory or the delimitation of any frontiers.

Widely ranging national stillbirth rates



Women from vulnerable and marginalized populations face a greater likelihood of having a stillborn child due to disparities in health care access

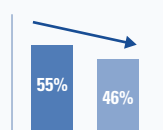
Stillbirths are more likely to occur among women with lower socio-economic status or lower levels of education and women living in rural or remote areas. Higher stillbirth risks are also seen when birth intervals are shorter and among the youngest mothers and the oldest mothers.



The proportion of stillbirths that occurs during labour – i.e., intrapartum stillbirths – is a critical piece of data for action because most stillbirths that occur during labour are preventable

Every 36 seconds, a baby is stillborn during labour. While the proportion of intrapartum stillbirths has declined since 2000 – from 55 per cent to **46 per cent** – stark inequities persist. In Europe, just 1 in 12 stillbirths occur during labour compared to **1 in 2** in sub-Saharan Africa and Southern Asia. These deaths are not inevitable. Likewise, many antepartum stillbirths are also preventable. When evidence-based interventions that improve the health of mothers and their babies are universally available and accessible along the continuum of care – such as quality antenatal care, skilled birth attendance, timely and continuous fetal monitoring, and emergency obstetric interventions – most of these babies could be saved.

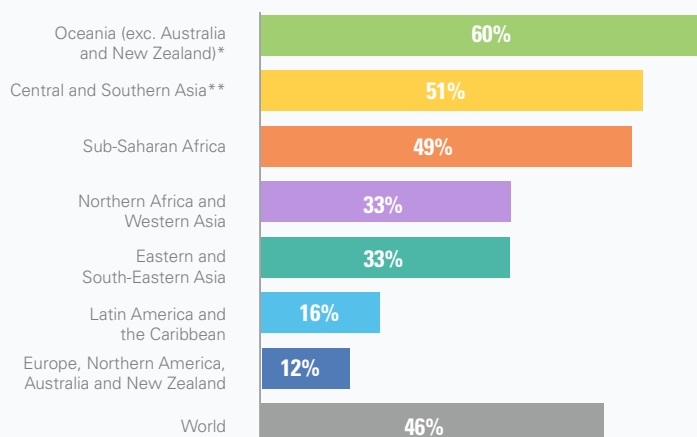
Closing these gaps requires urgent investment in equitable health systems and quality data on the timing of stillbirths to enhance national accountability.



The proportion of intrapartum stillbirths has declined since 2000 – from 55 per cent to **46 per cent**



1 in 2 stillbirths occurs during labour in sub-Saharan Africa and Southern Asia.



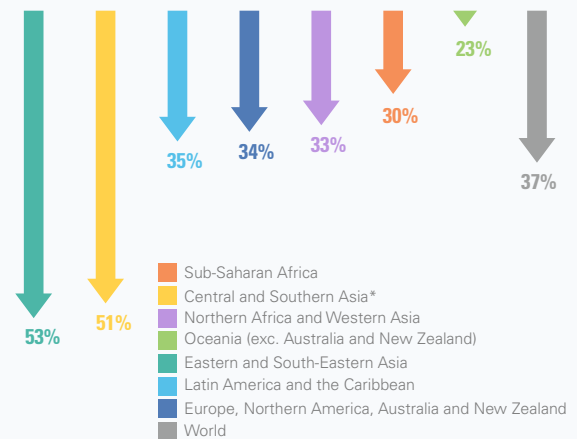
Note: *Given Oceania's (excluding Australia and New Zealand) small population and very limited availability of data, uncertainty around the region's estimates is large. **For Central and Southern Asia, the percentage of stillbirths that occurs during labour is 30 per cent for Central Asia and 51 per cent for Southern Asia.

Progress in reducing the stillbirth rate has been inconsistent and far too slow, especially when compared to other areas of child survival

The global stillbirth rate has declined by **37 per cent** since 2000, while the annual number of stillbirths has dropped by **39 per cent**. But the rate of reduction has slowed in the past decade: Between 2000 and 2015, the annual global rate of reduction was **2.5 per cent**, compared to just 1.1 per cent between 2015 and 2023.

Regional disparities in stillbirth prevention reveal widely varying levels of progress. Sub-Saharan Africa continues to have the world's highest stillbirth rate (22.2 per 1,000 births), but its reduction since 2000 has been just **30 per cent** – one of the lowest globally. In contrast, Eastern and South-Eastern Asia achieved the largest decline, reducing the stillbirth rate by **53 per cent**.

Global progress in reducing the stillbirth rate, however, is inadequate when compared to the more than 50 per cent decline achieved in under-five deaths since 2000. As a result of this uneven progress, the share of stillbirths among child deaths has grown. Unless further political and financial commitments are made to prioritize maternal and perinatal care, this gap will widen further.

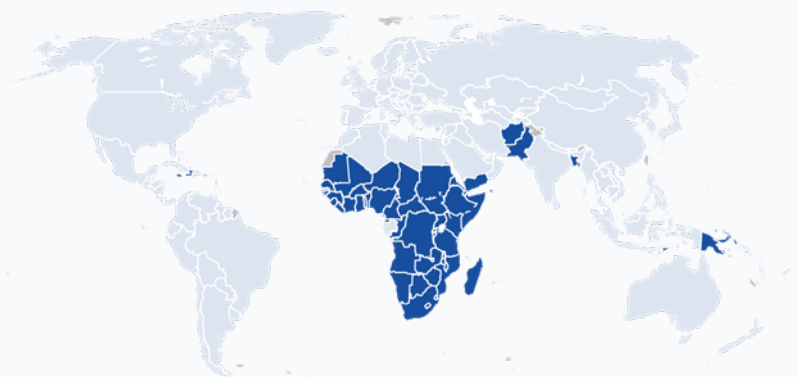


Note: *From 2000 to 2023, the percentage decline in the stillbirth rate was 51 per cent in Central and Southern Asia, as well as in both Central Asia and Southern Asia individually.

Without urgent action, the tragedy of stillbirth will needlessly impact millions of families

If current trends continue, **13 million more** babies will be stillborn by 2030, with half of these deaths in sub-Saharan Africa and nearly a third in Southern Asia. Additionally, **53 countries** will not meet the EWENE stillbirth target of 12 or fewer stillbirths per 1,000 total births by 2030, with **43** of these countries in sub-Saharan Africa. Thirty-two countries are projected to only achieve this target after 2050. According to EWENE, two thirds of reporting countries have not set a national stillbirth target.

The 53 countries that will not achieve the EWENE still-birth target, under current trends

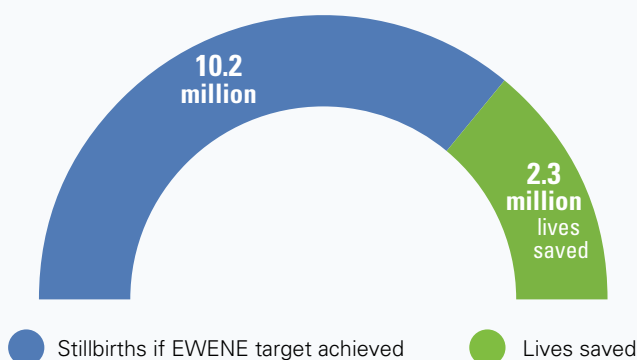


Note: Map does not reflect a position by UN IGME agencies on the legal status of any country or territory or the delimitation of any frontiers.

Achieving the target by 2030, however, could save 2.3 million babies. To meet global goals, 47 countries must at least double their current rate of progress in reducing stillbirths. Stronger investments, health care improvements and policy commitments must be made to protect these lives.

If all countries achieve the EWENE stillbirth target by 2030:

- **2.3 million** more babies will live: The number of stillbirths would drop from 12.5 million to **10.2 million**



Stillbirth is a life-changing event. Bereaved women must be met with respectful care in health care settings and by society and communities at large

Despite affecting millions of women every year, stillbirths remain nearly invisible in personal and public conversations, unaddressed across community health care systems, and neglected as a global public health issue. Women's lived experiences must be centred in policymaking and public discourse. Decision makers, community leaders and health care providers have a responsibility to actively listen, confront the systemic failures contributing to stillbirths, and ensure affected families receive trauma-informed care and equitable access to mental health services and psychosocial support.



Reductions to health financing threaten the progress that has been achieved in stillbirth prevention and are likely to jeopardize the lives of children.

Stillbirth is a neglected maternal and newborn health issue, with investments in this area lagging behind those that have been made in child survival. Any decrease in funding and support that impacts maternal and newborn health is likely to exacerbate the already concerning situation of stillbirth around the globe. A rise in the number of stillbirths is likely to be one of the first impacts if the provision of accessible, high-quality maternity services, including antenatal and childbirth care, is reduced.

Universal access to high-quality maternity care, improved stillbirth data, stronger policies, and increased awareness and support for affected families are the essential ingredients needed to prevent stillbirths and soften their effect on women and their communities. Change is possible, as seen in **28 countries** – including six low- and lower-middle-income countries – that cut their stillbirth rate by more than half from 2000 to 2023, with robust policy, investments and programmes playing a crucial role in these success stories. Renewed commitment and investment are urgently needed to end preventable stillbirths and address equity gaps among the most vulnerable women and children in every country around the world.

Box 1. UN IGME's approach to stillbirth estimation

Due to data availability and to allow for international comparison, the stillbirth rates and number of stillbirths in this report are standardized for the gestational age of 28 weeks or more (the third trimester of pregnancy).

While this report uses point estimates, these values include a 90 per cent uncertainty interval (see the Statistical Annex table). Estimates and analysis are based on data collected in 2024 for the years 2000 to 2023. For details on UN IGME's methodology, please see the Annex.

In order for the UN IGME to produce its stillbirth estimates every other year and adequately monitor this important maternal and newborn health concern, countries must collect quality data on every birth – whether

live or stillborn. The UN IGME urges national governments to improve stillbirth data and monitoring by leveraging and investing in existing data collection mechanisms, such as health management information systems (HMIS), sample vital registration systems and household surveys, to more accurately estimate how many babies are stillborn each year and target investment implementation of measures to reduce this number.

Only a limited number of high-burden countries provide reliable data using the 22-weeks-or-more threshold, as per the World Health Organization (WHO) International Classification of Diseases (ICD)-11 definition. If stillbirths at 22 weeks or more were included, the global burden would be approximately 30 per cent higher, with about half a million more stillbirths in 2023 alone.



QUICK REFERENCE ON STILLBIRTH

QUICK REFERENCE on stillbirth

Standardized definitions and reporting practices that align with international standards are the foundation of quality data on and monitoring of stillbirth. Policymakers rely on these data to make informed, effective decisions on the interventions needed to prevent stillbirths and provide care to mothers after stillbirths have occurred. This section presents a high-level overview of critical definitions, data collection needs and risk factors related to stillbirth. Health care workers and data managers seeking further guidance on definitions and criteria for collecting information when a stillbirth occurs can refer to the additional material available at <https://childmortality.org>.⁷

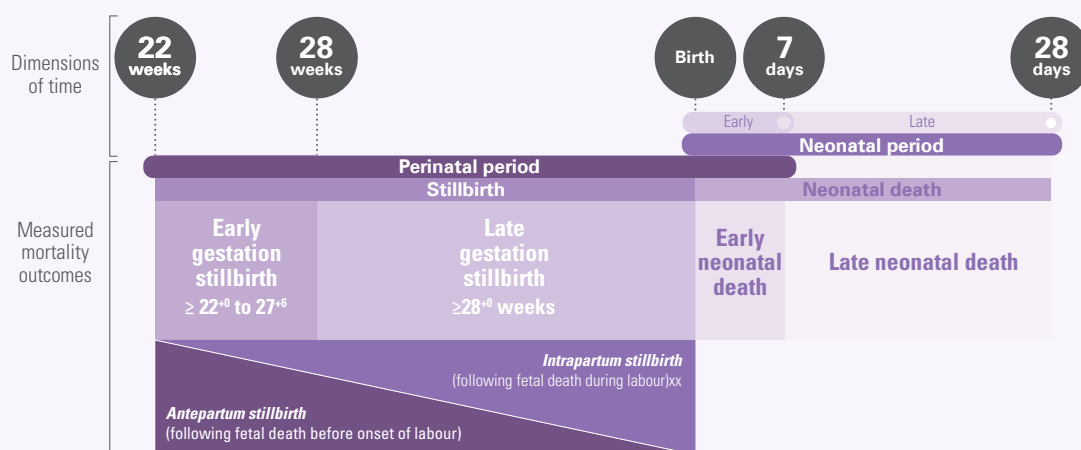
What is a stillbirth?

A stillbirth is a baby born with no signs of life at 22 or more completed weeks of gestation, as defined by the ICD 11th Revision.⁸ Stillbirths are categorized by age: early gestation stillbirth (at 22 to 27 completed weeks of gestation) and late gestation stillbirth (at 28 or more completed weeks of gestation) (see Figure 1). Due to low data availability and for purposes of international comparison, UN IGME stillbirth estimates represent late gestation stillbirths. Because early gestation stillbirths are not included, the true global burden of these deaths is approximately one-third greater.⁹

What is the difference between an antepartum stillbirth and an intrapartum stillbirth?

An antepartum stillbirth follows a fetal death that has occurred before the onset of labour. An intrapartum stillbirth follows a fetal death that has occurred during labour. Approximately 46 per cent of stillbirths occur intrapartum – most of these are preventable.

Figure 1: Definitions used in perinatal and neonatal mortality



What data are required for health systems to accurately measure stillbirth rates?

- ▶ For every birth outcome to be correctly classified, WHO recommends that health systems record a standard minimum perinatal dataset.
- ▶ This dataset includes vital status at birth (collected under details of death), gestational age and birthweight.
- ▶ Accurate measurement of these three key variables is an essential step to classifying a death as a stillbirth.
- ▶ Where possible, a cause of death for each stillbirth should be recorded using the international form, Medical Certification of Cause of Death.
- ▶ At minimum, all countries should collect information on the overall number and timing (ante partum or intra partum) of stillbirths, even if collecting detailed information on the cause of death is not possible at the time of the event.
- ▶ At institutional deliveries, labour room records should reflect the timing of a stillbirth.

What are the causes and risk factors associated with stillbirth?¹⁰

Preventing most stillbirths is possible. Many of the determinants associated with stillbirth can be addressed through sound policies and interventions – some, like malaria and syphilis, are treatable – while others – like household wealth and mother’s education – provide important information on the most vulnerable groups of women. The causes and risk factors tied to stillbirth must be identified and studied.



Access to health services

- Quality of antenatal care (number of visits during pregnancy)
- Quality of care during labour and childbirth



Environmental impacts

- Exposure to high temperatures
- Indoor and outdoor air pollution



Maternal fertility-related conditions

- Complications during birth (e.g., hypoxia)
- Ante partum haemorrhage (e.g., placental abruption)
- Fetal growth restriction
- Consanguinity
- Maternal age (younger than 16, older than 35)
- Short birth interval
- Infections and maternal conditions
 - Malaria, syphilis, HIV
 - Hypertensive disorders of pregnancy, diabetes, obesity



Behavioural factors

- Smoking and substance abuse



Socio-economic determinants

- Rural residence
- Migration status
- Experiences of sexual, physical, violent assault
- Ethnicity
- Economic status
- Maternal education



STILLBIRTH TODAY: LEVELS AND TRENDS



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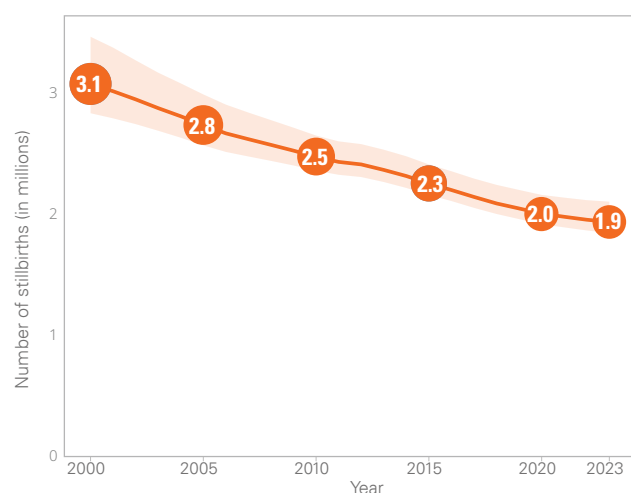
In 2023, an estimated 1.9 (1.8, 2.1)* million babies were stillborn at 28 weeks or more of gestation (see Figure 2) – down from 3.1 (2.9, 3.5) million in 2000. This constitutes 1 in 70 births ending in stillbirth, with one occurring every 17 seconds in 2023. The global

stillbirth rate in 2023 was 14.3 (13.7, 15.5) stillbirths per 1,000 total births – a reduction of 37 per cent since 2000 when it was 22.6 (20.9, 25.4) stillbirths per 1,000 total births.

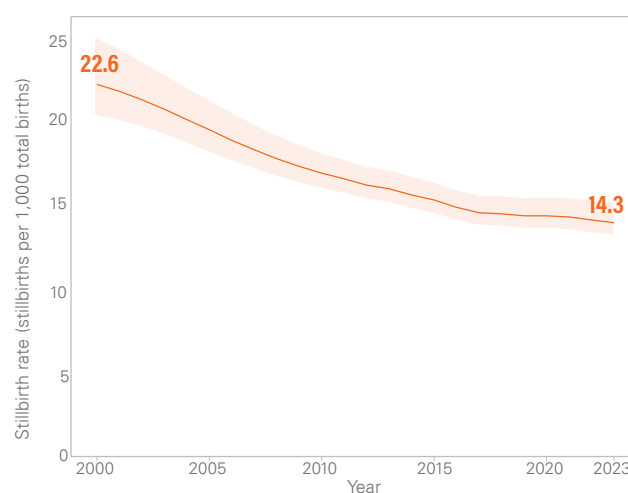
In 2023 alone, 1.9 million babies were stillborn

Figure 2: Number of stillbirths and global stillbirth rate (2000–2023)

2a) Number of stillbirths



2b) Stillbirth rate



Note: The solid line represents the median and the shaded area represents the 90 per cent uncertainty around the median value.

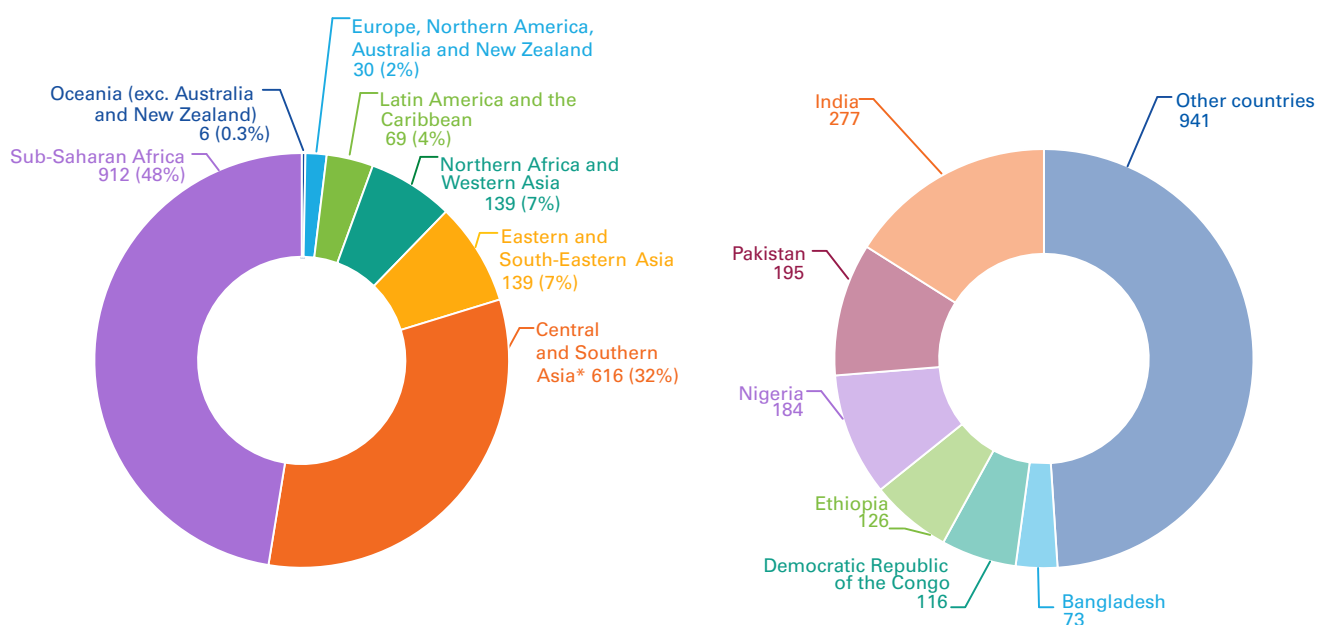
* Throughout this document, two values in parentheses indicate the 90 per cent uncertainty interval for the preceding number. Owing to the lack of availability of stillbirth data in some countries and regions, uncertainty intervals may be large and should be considered when interpreting point estimates. Uncertainty intervals for all stillbirth indicators are provided in the Statistical Table in the Annex. Rates greater than 10 are rounded to zero digits, and rates below 10 are rounded to one digit. Unrounded rates are available at <<https://childmortality.org>>.

Stillbirths remain concentrated in sub-Saharan Africa and Southern Asia, which together accounted for 79 per cent of stillbirths globally in 2023, a stark reflection of systemic health care inequities between regions. Sub-Saharan Africa alone bears nearly half (48 per cent) of the world's stillbirths despite representing only 30 per cent of live births, while Southern Asia accounts for nearly a third (32 per cent) of total stillbirths (see Figure 3). In 2023, more than a third of stillbirths (34 per cent) took place in three countries: India, Nigeria and Pakistan, which together accounted for 29 per cent of live births. Low- and lower-middle-income countries accounted

for 87 per cent of all stillbirths and 68 per cent of all live births in 2023 (low-income countries: 32 per cent of all stillbirths and 19 per cent of all live births; lower-middle-income countries: 56 per cent of all stillbirths and 49 per cent of all live births). In upper-middle- and high-income countries, disproportionately more babies are born alive. Countries in these income groups accounted for 13 per cent of the estimated stillbirths that occurred in 2023 and 32 per cent of live births (upper-middle-income countries: 11 per cent of all stillbirths and 22 per cent of all live births; high-income countries: 2 per cent of all stillbirths and 10 per cent of all live births).

Four in five estimated stillbirths occurred in two regions, while about half of the estimated stillbirths in 2023 were recorded in six countries

Figure 3: Stillbirths by Sustainable Development Goal region and country, in thousands (2023)



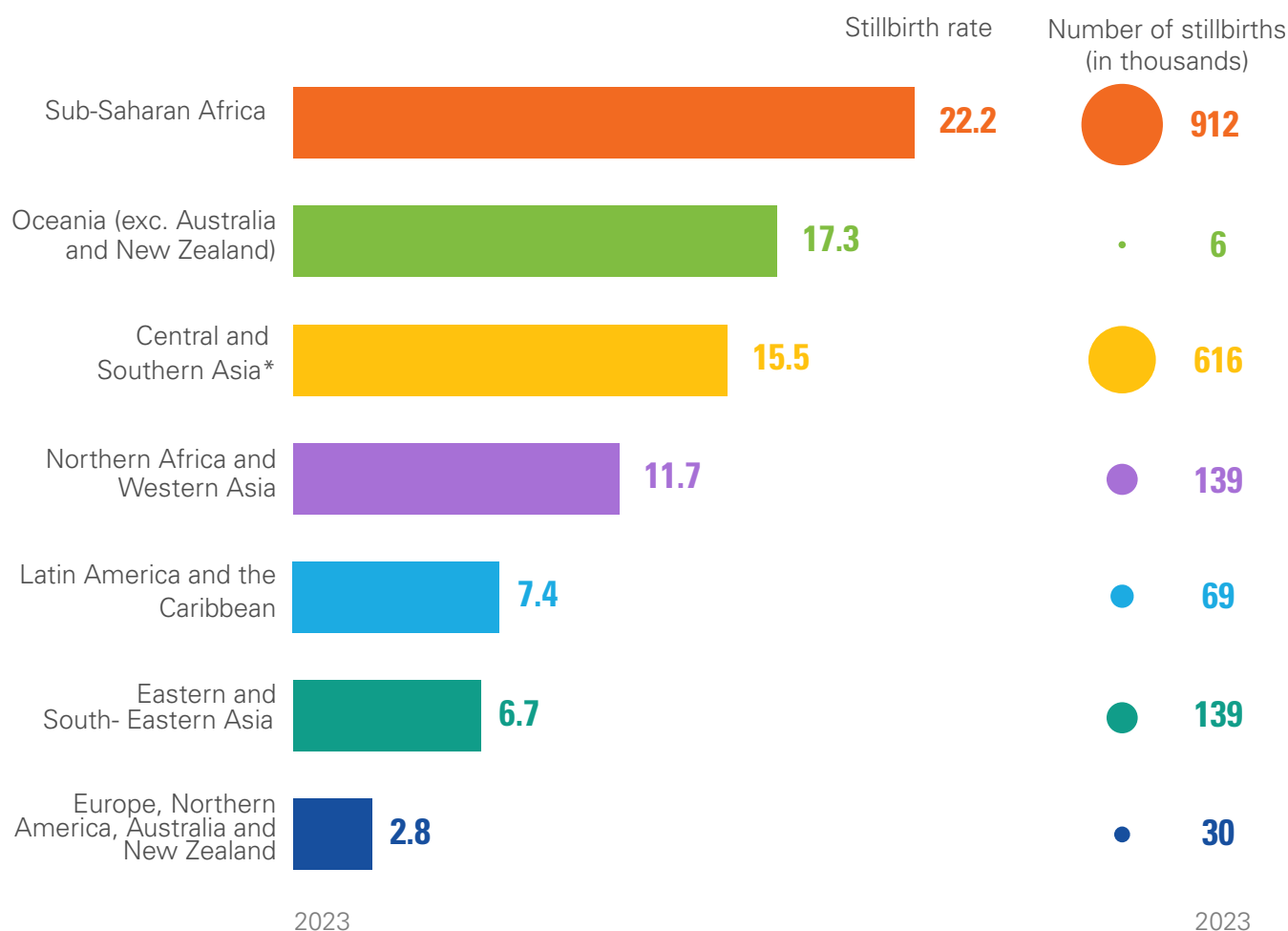
Note: *In 2023, Central and Southern Asia's total number of stillbirths was 616,000, accounting for 32 per cent of the global total. Of that, Central Asia's number of stillbirths was 13,000 (0.6 per cent of the global total), and Southern Asia's number of stillbirths was 604,000 (32 per cent of the global total). The number of stillbirths is determined by both the stillbirth rate and the total number of births in a country.

In sub-Saharan Africa, the estimated stillbirth rate of 22.2 stillbirths per 1,000 total births was 7.9 times the rate found in the region of Europe, Northern America, Australia and New Zealand, which had the lowest average stillbirth rate of 2.8 stillbirths per 1,000 total births. Oceania (excluding Australia and New Zealand) had the second-highest estimated stillbirth rate at 17.2 stillbirths per 1,000 total births. Southern Asia had the third-highest estimated rate, with 16.0 stillbirths per 1,000 total births in 2023 (see Figure 4).

In low-income countries, 1 in 43 births ended in a stillbirth compared to 1 in 356 in high-income countries. On average, the risk of stillbirth in low-income countries was 8.3 times that in high-income countries in 2023. The stillbirth rate in low-income countries was estimated to be 23.3 stillbirths per 1,000 total births in 2023, compared to a rate of 2.8 stillbirths per 1,000 total births in high-income countries.

Huge variations exist across regions

Figure 4: Stillbirth risk and number of stillbirths by Sustainable Development Goal region (2023)



Note: *For Central and Southern Asia, the stillbirth rate in 2023 was 15.5 stillbirths per 1,000 total births. Central Asia's stillbirth rate was 6.6 and Southern Asia's was 16.0. Central and Southern Asia's total number of stillbirths in 2023 was 616,000 (32 per cent of the global total). Of that, Central Asia's number of stillbirths was 13,000 (0.6 per cent of the global total), and Southern Asia's number of stillbirths was 604,000 (32 per cent of the global total).

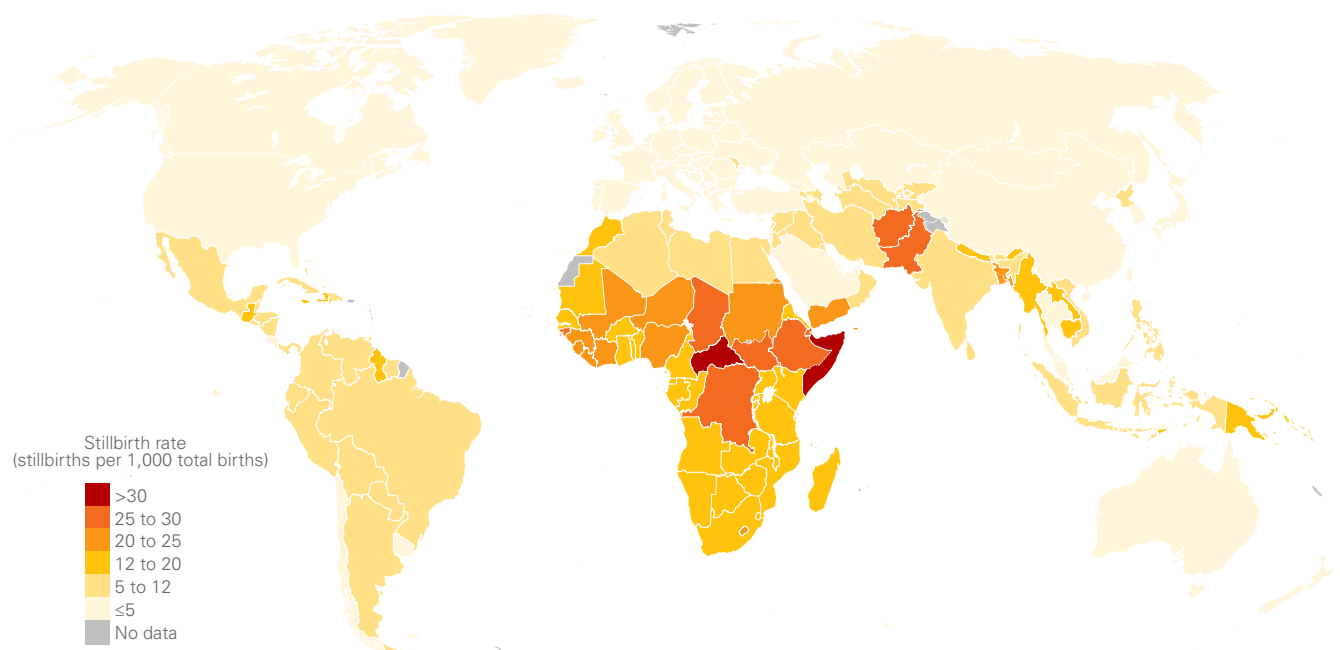
Newborn and child survival is significantly shaped by the place where a mother gives birth. In 2023, stillbirth rates across countries ranged from 1.7 stillbirths per 1,000 total births to 34.9 stillbirths per 1,000 total births (see Map 1 and Figure 5). In 10 countries, the stillbirth rate was greater than 25 stillbirths per 1,000 total births, and 58 countries have not reached the EWENE target of 12 stillbirths per 1,000 total births. Pregnant women in the country where the stillbirth rates were highest faced a risk of stillbirth that was about 21 times that of mothers in the country with the lowest rate among the 200 countries with estimates. The highest stillbirth rates were in countries in sub-Saharan Africa and Southern Asia and in the group of low-income countries (see Figure 5).



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Differences between countries facing stillbirth risks

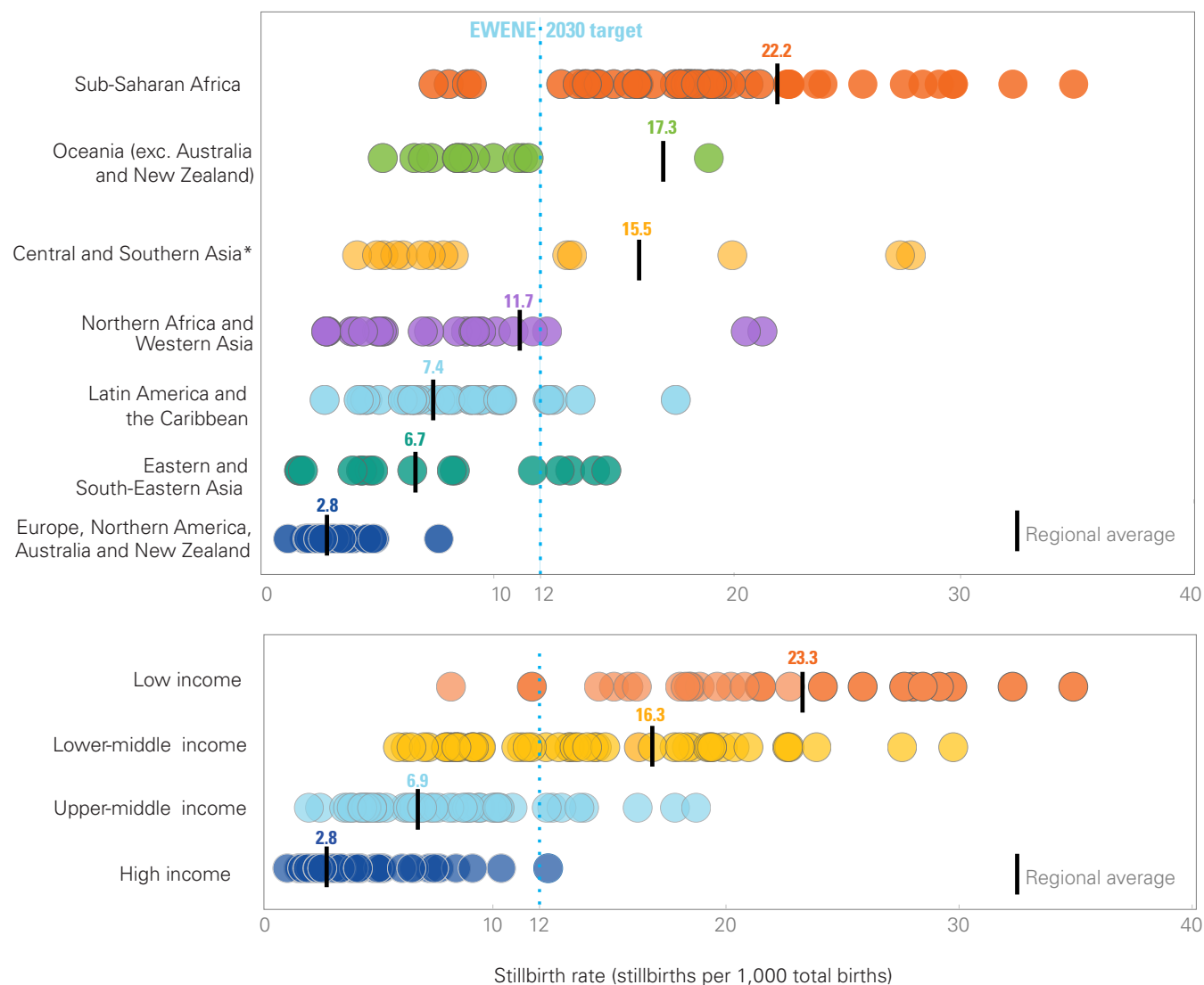
Map 1: Stillbirth rate, by country (2023)



Note: Categories are based on unrounded numbers; value ranges are greater than the lower bound number and less than or equal to the upper bound number. Map does not reflect a position by UN IGME agencies on the legal status of any country or territory or the delimitation of any frontiers.

Countries in sub-Saharan Africa and Southern Asia have the highest rates

Figure 5: Stillbirth rate by Sustainable Development Goal region and national income groups (2023)



Note: Dots represent individual countries in a region. Solid black lines represent regional averages. *Central and Southern Asia's average stillbirth rate in 2023 was 15.5; Central Asia's stillbirth rate was 6.6 and Southern Asia's was 16.0. National income classification follows the World Bank income classification, 2024. Details can be found at: <<https://datahelpdesk.worldbank.org/knowledgebase/articles/906519>>, accessed 23 July 2024. Among the 200 countries with stillbirth estimates, 26 are classified as low income, 51 as lower-middle income, 54 as upper-middle income and 65 as high income. Five countries/territories are not classified.

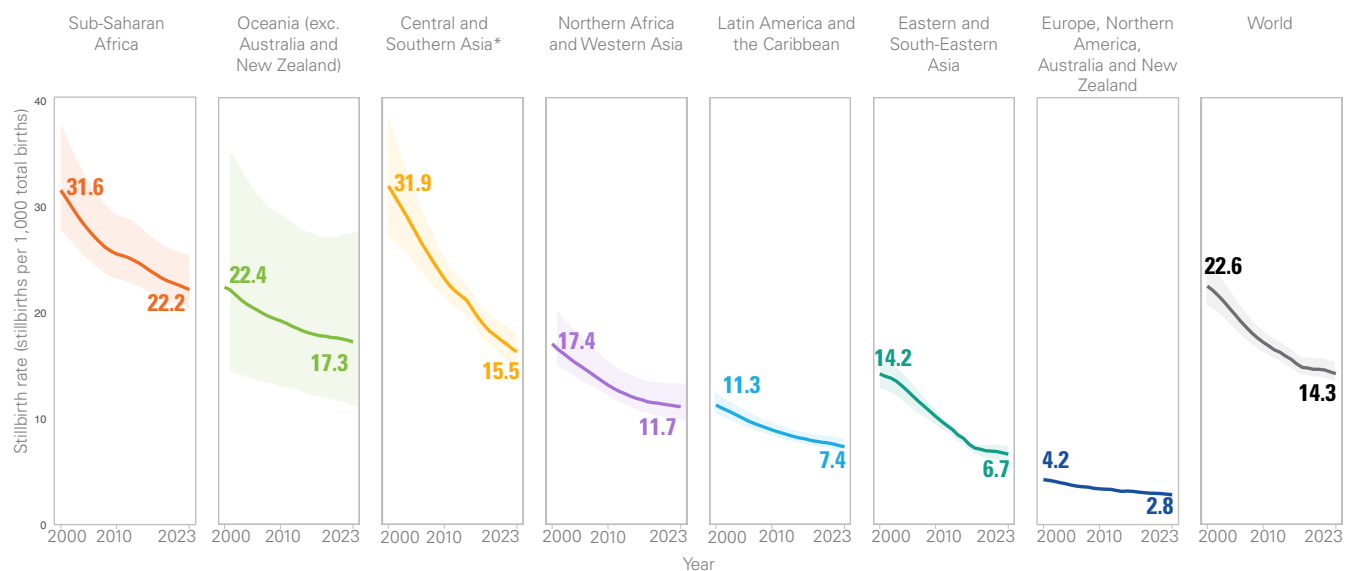
Between 2000 and 2023, the annual number of babies that were stillborn at 28 weeks or more of gestation declined by over 39 per cent, from 3.1 million to 1.9 million. At the same time, the global stillbirth rate decreased from 22.5 stillbirths per 1,000 total births in 2000 to 14.3 stillbirths per 1,000 total births in 2023 – a reduction of 37 per cent. But since 2015, progress has slowed: Between 2000 and 2015, the stillbirth rate declined by 2.5 per cent annually compared to just 1.1 per cent annually between 2015 and 2023.

A woman's risk of experiencing a stillbirth has decreased in every region (see Figure 6 and Statistical Table in the Annex). The most significant reductions were estimated in Eastern and South-Eastern Asia, where the rate declined by 53 per cent from 14.2 stillbirths per 1,000 total births in 2000 to 6.7 stillbirths per 1,000 total births in 2023, and in Central and Southern Asia, which saw a 51 per cent reduction from 31.9 stillbirths per 1,000 total births in 2000 to 15.5

stillbirths per 1,000 total births in 2023. Progress was much slower in sub-Saharan Africa, where the stillbirth rate declined by 30 per cent between 2000 and 2023, from 31.6 stillbirths per 1,000 total births to 22.2 stillbirths per 1,000 total births. Regions with already low stillbirth rate levels – Latin America and the Caribbean and Europe, Northern America, Australia and New Zealand – had more modest declines.

Stillbirth risk remains high in sub-Saharan Africa

Figure 6: Stillbirth rate, by Sustainable Development Goal region (2000–2023)



Note: The solid line represents the median and the shaded area represents the 90 per cent uncertainty around the median value. *In 2000, the combined stillbirth rate for Central and Southern Asia was 31.9 stillbirths per 1,000 total births; Central Asia's stillbirth rate was 13.4 and Southern Asia's was 32.4. In 2023, the region's combined stillbirth rate was 15.5; individually, Central Asia's stillbirth rate was 6.6 and Southern Asia's was 16.0.



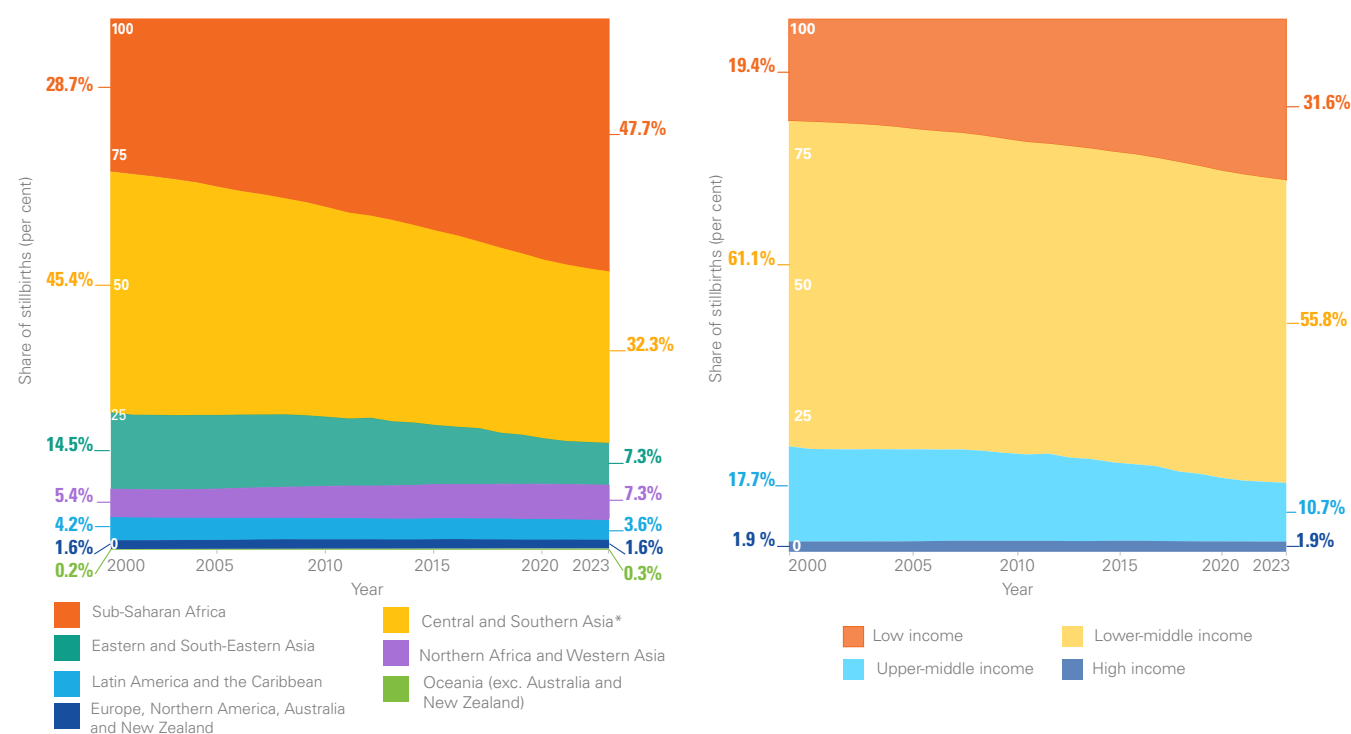
In sub-Saharan Africa, the stillbirth burden carried by families is growing (see Figure 7). In 2000, the annual number of stillbirths stood at 897,000, and in 2023, it increased to 912,000. The proportion of global stillbirths occurring in sub-Saharan Africa has risen sharply – from 29 per cent in 2000 to 48 per cent in 2023 (see Figure 7). Population shifts across several sub-Saharan countries are driving this trend, where total births have grown at a faster rate than the decline in stillbirth rates. Between 2000 and 2023, total births in the region surged by 45 per cent, while global birth rates remained steady. In contrast, the number of stillbirths in Central

and Southern Asia dropped from over 1.4 million in 2000 to 616,000 in 2023, due to more rapid reduction in stillbirth rates, coupled with fertility declines. Across all other regions except Oceania (excluding Australia and New Zealand), stillbirths have declined.

The growing number of stillbirths that occur in low-income countries is rising, where stillbirths reached 551,000 in 2023 – up from 496,000 in 2000. Countries in this income group now bear a greater share of the global burden, with their portion increasing from 19 per cent in 2000 to 32 per cent in 2023.

Stillbirths are increasingly concentrated in sub-Saharan Africa and in low-income countries

Figure 7: Share of stillbirths, by Sustainable Development Goal region and national income groups (2000–2023)

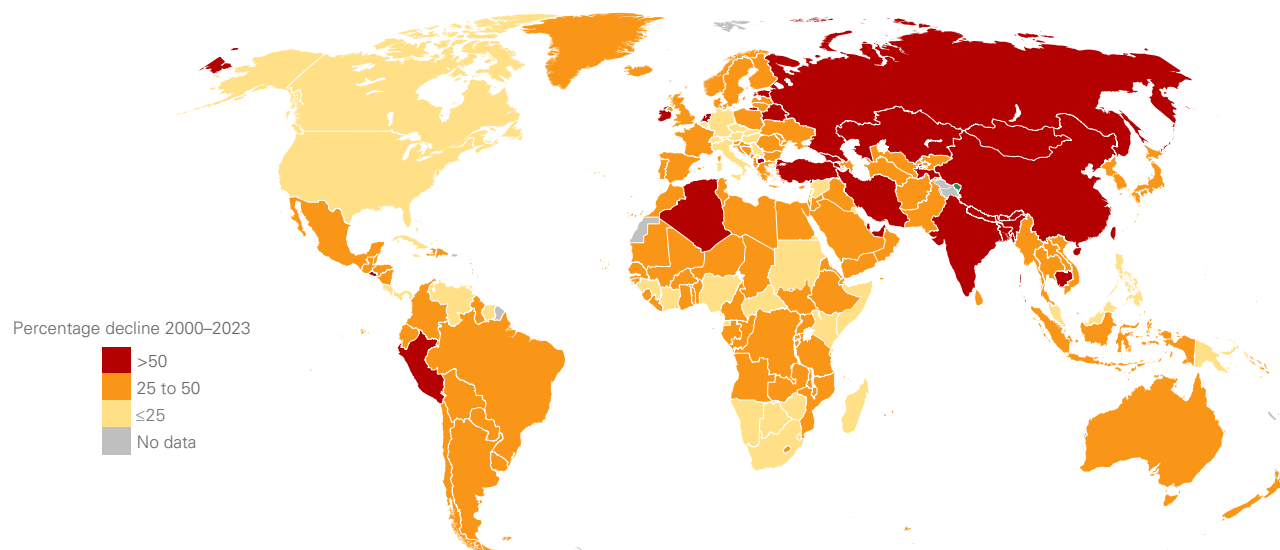


Note: *Of Central and Southern Asia's 45.4 per cent share of global stillbirths in 2000, Central Asia accounted for 0.5 per cent and Southern Asia for 44.9 per cent; of the region's 32.3 per cent share of global stillbirths in 2023, Central Asia accounted for 0.7 per cent and Southern Asia for 31.6 per cent. National income classifications follow the World Bank classification, 2024. Among the 200 countries with stillbirth estimates, 26 are classified as low income, 51 as lower-middle income, 54 as upper-middle income and 65 as high income. Five countries/territories are not classified. Note that countries follow the same classification through all years in this chart.

Several countries, including some that are resource-poor, have made encouraging gains in reducing their stillbirth rate. A total of 28 countries – including 6 lower-middle-income countries (Cambodia, Bhutan, Nepal,

India, Tajikistan and Bangladesh) – have reduced their stillbirth rate by more than half (see Figure 8, Figure 9 and Map 2). No country in sub-Saharan Africa reduced its stillbirth rate by more than 25 per cent (see Map 2).

Map 2: Percentage decline in stillbirth rate, by country (2000–2023)



Note: Categories are based on unrounded numbers; value ranges are greater than the lower bound number and less than or equal to the upper bound number. Map does not reflect a position by UN IGME agencies on the legal status of any country or territory or the delimitation of any frontiers.

Progress is possible

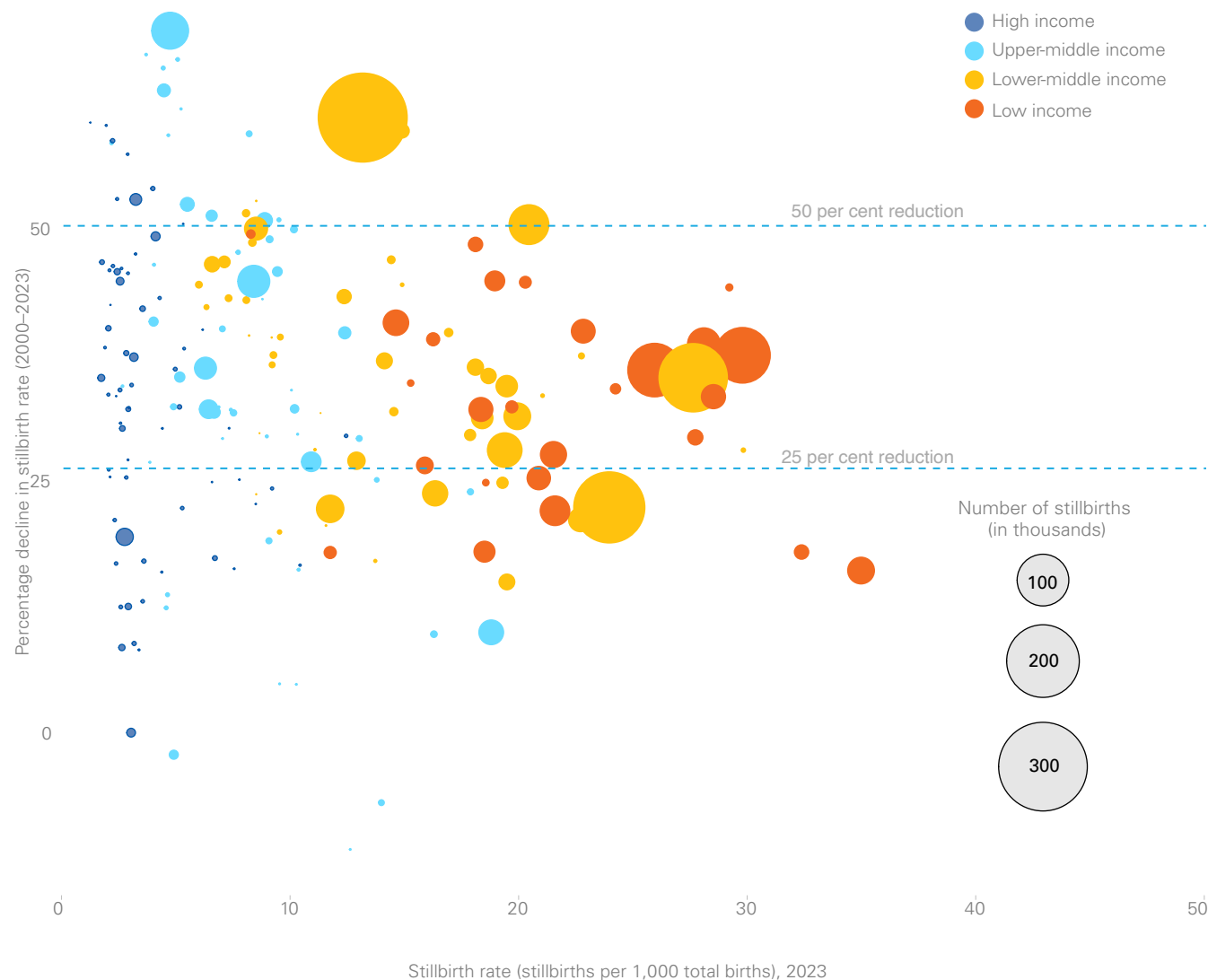
Figure 8: Countries with over a 50 per cent decline in the stillbirth rate (2000–2023)

	Kazakhstan	70 (60, 77)%		Netherlands	58 (53, 63)%
	China	69 (61, 76)%		Belarus	58 (44, 69)%
	North Macedonia	67 (60, 72)%		Qatar	57 (49, 64)%
	Georgia	66 (57, 74)%		United Arab Emirates	54 (47, 60)%
	Mongolia	66 (57, 73)%		Ireland	53 (44, 60)%
	Türkiye	63 (54, 71)%		Russian Federation	53 (40, 62)%
	India	60 (48, 70)%		Iran (Islamic Republic of)	52 (28, 68)%
	Estonia	60 (51, 67)%		Tajikistan	51 (24, 69)%
	Cambodia	59 (37, 74)%		Peru	51 (32, 65)%
	El Salvador	59 (39, 73)%		Armenia	51 (33, 64)%
	Nepal	59 (42, 71)%		Algeria	51 (39, 60)%
	Kosovo (UNSCR 1244)	59 (39, 73)%		Bangladesh	50 (36, 61)%

Note: Values in parentheses indicate the 90 per cent uncertainty intervals for the percentage declines in the stillbirth rate between 2000 and 2023. Countries listed are selected based on percentage decline point estimates and exclude countries with no quality data on stillbirths and countries with less than 10,000 estimated live births in 2023. Additional countries that were top performers in reducing the stillbirth rate between 2000 and 2023 with less than 10,000 live births are Maldives, Monaco, Bhutan and Nauru. Monaco and Nauru have no quality data available. For some countries, estimates are derived from the covariate-driven model. In Map 2, categories are based on unrounded numbers; value ranges are greater than the lower bound number and less than or equal to the upper bound number. Map does not reflect a position by UN IGME agencies on the legal status of any country or territory or the delimitation of any frontiers.

Slow progress in lower income countries, and stillbirth rates remain high

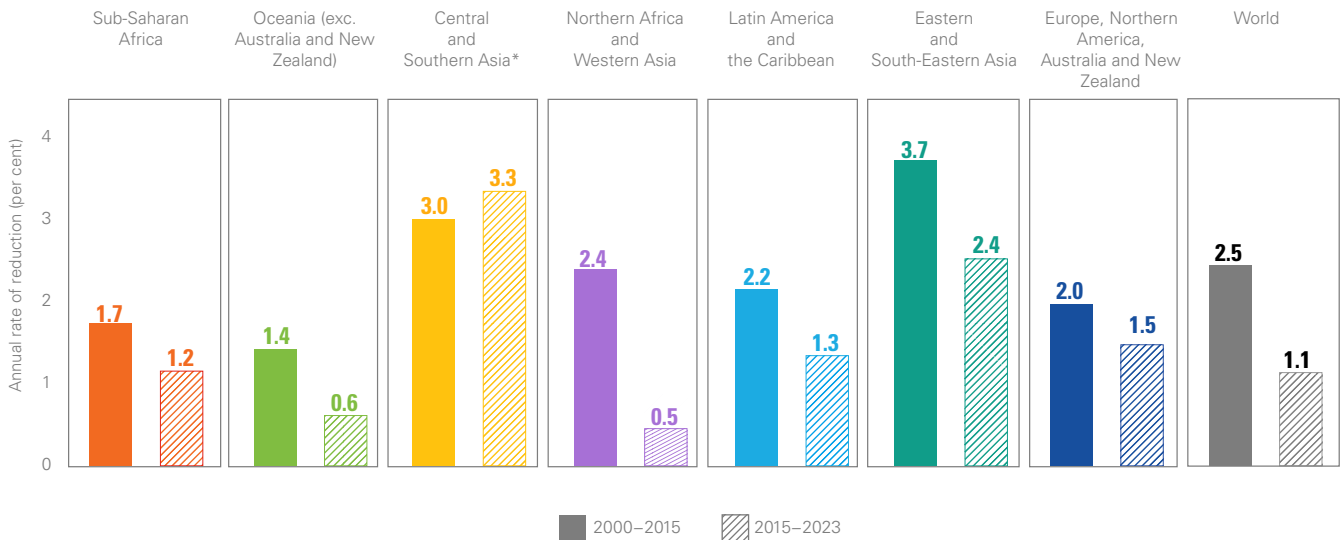
Figure 9: Percentage decline in stillbirth rate in 2000–2023 against stillbirth rate in 2023, by national income group



Note: National income classifications follow the World Bank classification, 2024. Among the 200 countries with stillbirth estimates, 26 are classified as low income, 51 as lower-middle income, 54 as upper-middle income and 65 as high income. Five countries/territories are not classified. Note that countries follow the same classification through all years in this chart.

Slower or stagnating progress since 2015 was observed across all regions

Figure 10: Annual rate of reduction in stillbirth rate, by Sustainable Development Goal region (2000–2015 and 2015–2023)



Note: *The annual rate of reduction for Central and Southern Asia from 2000 to 2015 was 3.0; in Central Asia, it was 3.3, and in Southern Asia, it was 2.9. From 2015 to 2023, the annual rate of reduction in Central and Southern Asia was 3.3; in Central Asia, it was 2.8, and in Southern Asia, it was 3.3.

Progress towards ending preventable stillbirths has faltered since 2015 (see Figure 10). Globally, the annual rate of reduction fell by more than half, from 2.5 per cent in 2000 to 2015 to 1.1 per cent between 2015 and 2023, with only Southern Asia showing acceleration. Southern Asia's annual rate of reduction of 3.3 per cent from 2015 to 2023, however, was still slower than the 3.7 per cent reduction achieved in Eastern and South-Eastern Asia from 2000 to 2015. The region with the highest stillbirth risk – sub-Saharan Africa – also had low progress rates.

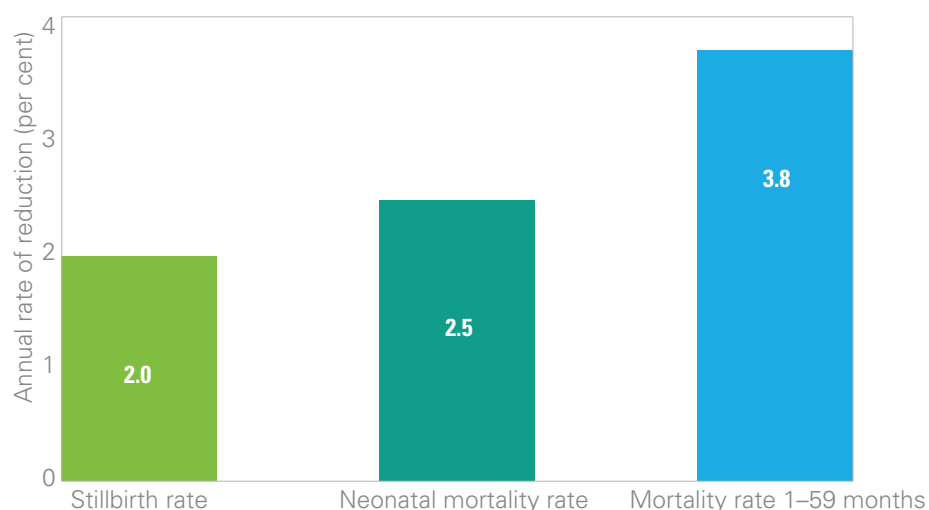
Progress in reducing the stillbirth rate has been slow (see Figure 11). The annual rate of reduction in the stillbirth rate was 2.0 per cent from 2000 to 2023, lower than the 2.5 per cent rate of reduction in neonatal mortality and the 3.8 per cent rate of reduction in mortality among children aged 1–59 months over the same period. This trend was observed in all regions but appears most dramatically in sub-Saharan Africa, where the annual rate of reduction in the stillbirth rate has remained below 2 per cent since 2000. The region's annual rate of reduction in mortality among children aged 1–59 months was triple that of its progress in

reducing the stillbirth rate (4.3 per cent versus 1.5 per cent). The result is that stillbirths account for a larger share of babies and children who die before turning 5.



Slower progress than in child survival

Figure 11: Annual rate of reduction in stillbirth rate and mortality rates among neonates and children aged 1–59 months globally (2000–2023)

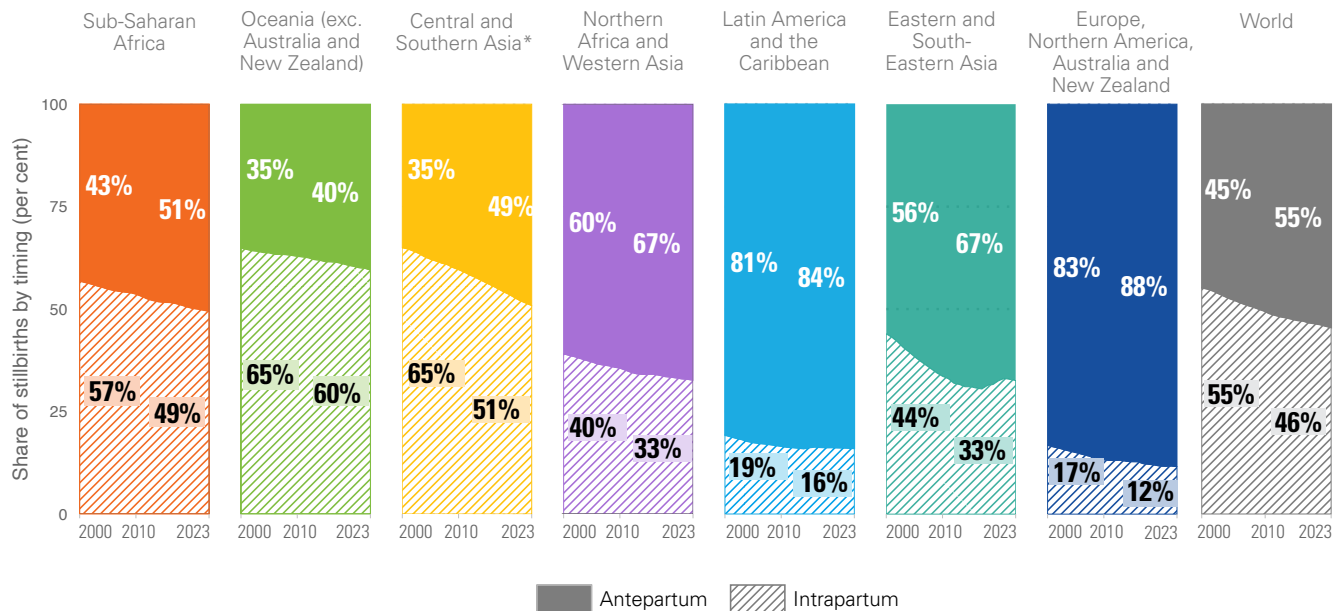


In 2023, there were nearly 900,000 intrapartum stillbirths – accounting for 46 per cent of all stillbirths (see Figure 12). Globally, the estimated intrapartum stillbirth rate stood at 6.5 stillbirths per 1,000 total births, meaning 1 in 153 babies died during delivery. The proportion of stillbirths that occur during labour is estimated to be 49 per cent in sub-Saharan Africa, 51 per cent in Southern Asia and 60 per cent in Oceania (excluding Australia and New Zealand) – amounting to almost 800,000 intrapartum stillbirths in the three regions combined and accounting for 87 per cent of all intrapartum stillbirths. In contrast, just 12 per cent of stillbirths in Europe, Northern America, Australia and New Zealand – or 4,000 stillbirths – occurred during labour. Globally, the number of intrapartum stillbirths has declined by almost half, from 1.7 million in 2000 to below 900,000 in 2023, with the share declining from 55 per cent of all stillbirths to 46 per cent. Though the proportion of intrapartum stillbirths has shrunk across all regions since 2000, further efforts are needed to accelerate progress, particularly in the regions where intrapartum stillbirth rates and the burden of stillbirths remain the highest.



Almost half (46 per cent) of stillbirths occur during labour – and are mostly preventable

Figure 12: Proportion of antepartum and intrapartum stillbirths during, by Sustainable Development Goal region (2000–2023)



Note: *In 2000, the share of intrapartum stillbirths in Central and Southern Asia was 65 per cent; in Central Asia, intrapartum stillbirths accounted for 44 per cent of all stillbirths, and in Southern Asia, they accounted for 65 per cent. In 2023, the combined intrapartum share in Central and Southern Asia was 51 per cent; in Central Asia, intrapartum stillbirths accounted for 30 per cent of all stillbirths, and in Southern Asia, they accounted for 51 per cent.





A NEED TO GRIEVE, A RIGHT TO BE HONoured

For many women, losing a baby due to stillbirth is the beginning of a long journey of recovery. Though the emotional wounds may be hard to see, they often profoundly impact the well-being of bereaved mothers for years to come, affecting their mental and physical health, relationships, community ties, and financial situations, among other areas of day-to-day life. But social and cultural taboos, stigma and misconceptions about stillbirth leave many women without the answers and empathy they deserve. Compassionate bereavement care can take many forms – from being treated with respect and dignity by health care practitioners to honouring and memorializing a stillborn child.

In this section, bereaved mothers from around the world share their accounts of loss, grief and healing. Their stories leave little doubt that their stillborn sons and daughters will never leave their hearts.

Nonkululeko and Ntando South Africa



Courtesy of Nonkululeko Shibula

After my daughter was stillborn, a simple funeral was arranged by our relatives. No other services were held, based on the belief that acknowledging her might bring harm to any future children I might have. I longed to speak about her openly and express my sorrow, but I remained silent. The absence of rituals or spaces to remember Ntando left a void in my heart.

I found creative ways to honour and heal. I wrote poems and letters and drew pictures and painted. This helped me process and heal the trauma.

When stillbirth occurred again within my family, I recognized the familiar, silent suffering. This deepened my understanding of the pain so many parents endure, yet often in isolation. But healing looks and feels different for all of us.

Now, I work with families in my community who have experienced stillbirth to embrace their traditions and find new ways of remembrance. Honouring a child's existence does not have to conflict with cultural beliefs – it can be an extension of love, healing, and connection. No bereaved parent should feel alone, silenced or unable to acknowledge the profound impact of their child's brief but meaningful existence.



Courtesy of Nonkululeko Shibula

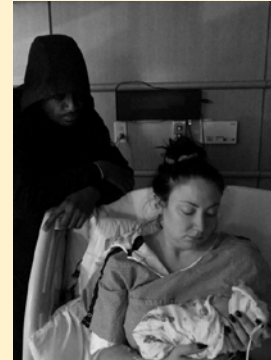
Elizabeth and Aaliyah

United States

When Aaliyah was born still, grief reshaped everything – my body, my mind, my sense of purpose. Caring for myself became a way to keep my baby close, a way to acknowledge that even in her absence, I was still her mother.

In the early days, that care looked like the simplest acts: drinking a glass of water, going outside for a breath of fresh air, wrapping myself in a blanket and allowing my body to rest. Treating myself with gentleness became an act of defiance against the guilt and anger I carried.

Movement has been a major form of healing. I started with gentle stretching in my home. Over time, it became a more intentional yoga practice in Aaliyah's room. In the rawest days of grief, eating felt impossible. But over time, food became a reminder that I deserved to be nurtured. Cooking became an unexpected form of therapy. Perhaps the most profound act of self-care was giving myself permission to feel everything – the deep sadness, love, longing, and even moments of joy. I learned that grief wasn't just about missing her; it was about carrying her forward with me.



Courtesy of Elizabeth O'Donnell

Yana and Fauzan

Indonesia



Courtesy of Nur Yanayirah

Come on, baby. Please wake up. I can't believe this happened. I love you. I don't want to lose you.' But it was true – my baby was dead. The doctor suggested that I be induced in the hospital. I just wanted the process to be quick.

The delivery room consisted of three small beds and various surgical equipment. It was a shared room with other grim-faced mothers whose babies had died. The lack of awareness and empathy from the health workers who seemed to blame me for my baby's death made my condition even worse. The midwife did not allow me to see, kiss, hug or even say goodbye to my baby – Muhammad Fauzan.

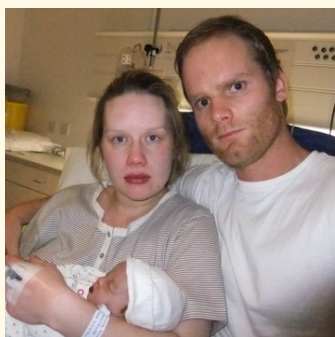
She was afraid that I would become hysterical and lose control. I was moved to another room with a mother who gave birth to a healthy baby. There was laughter, joy and cries of a baby, while I was alone and crying.

The trauma of giving birth to Fauzan continued to haunt me, even after I had given birth to a healthy baby girl a year later. Eventually, I was diagnosed with postpartum depression (PPD) and post-traumatic stress disorder. After undergoing treatment with a clinical psychologist and psychiatrist, I recovered from PPD.

I went on to create MotherHope Indonesia to provide support for bereaved parents and women affected by childbirth trauma and PPD. To honour Fauzan, I wrote several books about him and advocated to the Ministry of Health to expand post-miscarriage services to include stillbirth prevention and bereavement care services in Indonesia.

Nicole and Jessica

United Kingdom



Courtesy of Nicole Regan-White

Jessica was born on 24 December, weighing 5 pounds 10 ounces. I didn't want to see her at first. I was scared of what she would look like. "She's beautiful, she's perfect, and you will regret it if you don't meet her," the midwife told us. In the precious couple of hours we had with her, we held her and said hello and goodbye all in one breath. We took photos, saved a locket of her hair and made prints of her hands and feet.

At the hospital, we were handed a small bundle of leaflets. However, we could barely read and take in a sentence. Most of the material was lengthy, repetitive and out of date.

During pregnancy, we had been warned about the risk of sickle cell anaemia and Down syndrome, which were 1 in 2000 and 1 in 700 risk. However, we had never been told about the risk of stillbirth, which was 1 in 200 births in the UK, and out of 36 developed countries, the UK was 35th.

We remember Jessica every year by giving a birthday present 'from the stars' to her little brother and sister and lighting her birthday candle on Christmas Eve. We have never shied away from talking about Jessica to our children, and when we look up at the stars, we all know that she is shining back at us.

Talitha, Hosea and Malachi

United Arab Emirates

My tattoos mark me as Hosea and Malachi's mom. They have opened up opportunities to share my story unsuspecting passersby, which then always leads first to shock, pity or sympathy, and many times, to sharing their own stories of baby loss. Children have pointed out the feet on my arms, prompting discussion from their parents about loss, many times their own.

As we experienced our loss and have reflected upon the care we received, there are clear areas in which we have been well resourced, and others where there is still a lack. Bereavement care should, at minimum, include properly trained staff that respects the personhood of each baby, dead or alive; memory-making facilities – a dedicated room, cuddle cots, memory boxes, and printing or stamping supplies; and signposting to bereavement care groups and resources, which is perhaps the most simple yet impactful thing a clinical facility can do.



Courtesy of Nathaniel Daniel

When our sons died, the facility provided footprints, hair, clothing, tags, framed photos, and a room to take it all in without limits – all things we didn't know to ask for but are now incredibly impactful.

Sian and Lottie

United Kingdom

Our daughter had gone, but her body remained. The prospect of a funeral was not a welcome one. Planning Lottie's funeral felt painful and unnecessary. However, in those moments, our funeral director understood without explanation and arranged the entire process. She created a moment in time for myself and my husband to say a final goodbye to our daughter that we didn't know we needed.

Lottie's ashes posed another impossible decision. I couldn't bear to give her up to the world and scatter her ashes. For over a year, she existed in the small, white cardboard box we brought her home in, resting on a shelf in our living room in the heart of our home with the rest of her family. Eventually, I made the decision to wrap her ashes within the only teddy bear we bought for her before her death. We could all embrace her when we needed to.



Courtesy of Sian Reece

I also wrote her a letter:

To my beautiful Lottie,

I'm writing these words to you long before you could have ever spoken, read or understood them, but in the desperate hope that you'll hear them. The nine months we spent together were incredible. We loved watching my belly grow every day as you grew inside me. Your constant little kicks and wiggles kept me company every minute of every day and filled me with more love and happiness than you could imagine. Carrying you and holding you were the easiest things in the world.

Your daddy and I, and all your big sisters and brother, couldn't wait to meet you. We talked about you every day and dreamed about the day you would finally be in our arms and the future we would have together. I am so terribly sorry and so completely heartbroken that your time with us was too short. I'm sorry for any role I may have played in that and know that I would do anything to have you back in my arms. Lottie, I love you with all my heart, more than you could ever know, and I will carry you in my heart every day wherever I may go. I miss you so very much. This letter is the most difficult I have ever had to write, and no words can express just how much you are loved, my baby girl.

Sweet dreams my beautiful Lottie.

*All my love, always,
Mummy*

Katherine and Helen United States



*Mr. and Mrs. David Hensley
Announce the birth of their beloved daughter
Helen Louise
Sunday, November Sixth
Two Thousand and Five
Baptized into God's family and held in the arms of her family
She was received into the kingdom of heaven.
5 pounds 3 ounces 19 inches*

Courtesy of Katherine Hyde Hensley

Three days before my scheduled C-section, I learned that there was no heartbeat. Helen was born at 5 pounds and 6 ounces and looked just like her siblings.

After the funeral, the community returned to life. But our family faced stigma and isolation as a result of Helen's death. I struggled with milk supply while grieving, working and parenting two small children. I experienced severe postpartum depression, anxiety and PTSD and found it challenging to access mental health support, even from my obstetrician.

Helen's death blindsided our family. We learned about the frequency, prevalence and painful stigma and isolation that comes with infant death. Losing our daughter impacted our marriage, finances and attachment parenting style.

I have now made it my life's mission to raise awareness about the crucial importance of monitoring fetal movement, advocating maternal and infant mental health, and supporting birthers through the dark days of grief. Through education, advocacy, and empowerment, I strive to bring about positive transformations in the lives of bereaved parents, instilling hope and resilience.

Trilakshi and her firstborn

Sri Lanka

Although I have three children now, I can never forget my first baby. For a long time, I wept in a corner of my house, refusing food and drink, weeping. For a long time, I could feel the baby's tiny hands and body moving inside my belly. But due to the support of my family and the counselling of the health team, I recovered bit by bit.

I feel that if that baby was also here, all my children would have played together – been with me, with all of us, together. I write about the baby, do meritorious acts and offer merit to the baby's soul.

To my hope!

*Secretly, in a dream, you walk in,
Appearing faintly, then fading away.
Do you come and go, unseen?
Or are you here to stay? I don't know.*

*Your softest face touched by my hand,
I kiss it gently in my dreams.
A butterfly's delicate flutter I feel,
Remembering it again and again.
Do you come and go, unseen?
Or are you here to stay? I don't know...*

*At the feet of mother Paththini,
My tears fall like pearls, one by one.
Waiting for you forever,
With a heart that aches and trembles...
Do you come and go, unseen?
Or are you here to stay? I don't know...*

The need for respectful bereavement care for parents of stillborn children took centre stage at the International Stillbirth Alliance's recent conference in Colombo, Sri Lanka. The event culminated in a memorial service held in Independence Square that included religious rituals,

blessings, prayers and ceremonies conducted by leading priests of Sri Lanka's four major religions Buddhism, Hinduism, Catholicism and Islam. The service concluded with the symbolic lighting of candles by all participants, symbolizing hope and remembrance, healing and hope.



Courtesy of College of Community Physicians of Sri Lanka



PREDICTING AND PREVENTING TOMORROW'S TRAGEDIES

If stillbirth rates remain unchanged in every country and no further progress is made, 13.7 million babies will be stillborn between 2024 to 2030. Maintaining current rates of progress across all countries would reduce this to 12.5 million babies affected. If, however, all countries were to accelerate efforts to meet the EWENE target by 2030, the number could be further reduced to 10.2 million – or 2.3 million fewer lives lost than if current trends continue (see Figure 13). About 7 in 10 of these averted stillbirths would affect mothers and families in sub-Saharan Africa. Even at rates of 12 stillbirths per 1,000, however, many stillbirths are still preventable, as evidenced by the low stillbirth rates across high-income countries. If the gap were closed and every country reduced its rate to that of current high-income countries' rates of 2.8 stillbirths per 1,000 total births, 5.6 million babies would die between 2024 and 2030 – 4.6 million fewer stillbirths.

Aiming for an ambitious rate of progress – in which all countries perform at the 99th percentile of the country-level annual rate of reduction from 2000 to 2023 – 11 million babies would be stillborn from 2024 to 2030, which is a result similar to reaching the

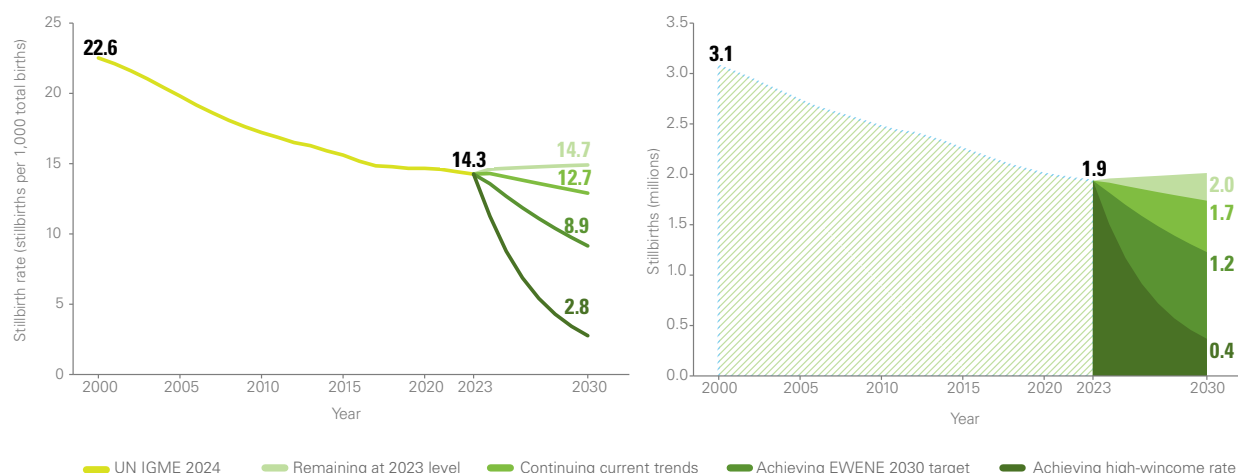
EWENE target in 2030. But if all countries perform at the 1st percentile of the country-level annual rate of reduction from 2000 to 2023, 14 million babies will be stillborn – a result similar to the constant no progress scenario.

The EWENE partnership builds on the legacy of ENAP and EPMM. It sets clear and ambitious targets to reduce maternal and newborn mortality and stillbirths by 2030, and it calls on countries to achieve a rate of 12 stillbirths or fewer per 1,000 total births by 2030.

Reaching the EWENE target would profoundly shape the futures of millions of women and their babies.

Achieving the EWENE target will save millions of babies

Figure 13: Estimated and projected global stillbirth rates and number of stillbirths, by scenario (2000–2030)



The EWENE stillbirth target will not be met by 2030 if current trends continue. Among the 200 countries studied, 139 have already achieved the target, and a further 8 are on track to meet it (see Map 3). Without urgent and rapid investments in health care systems, 53 countries will miss the target, 43 of which are in sub-Saharan Africa. More than half (32) of these 53 countries will not meet the target by 2050; the majority (26) are in sub-Saharan Africa. Unless current levels of funding and support are maintained, the progress that has been achieved since 2000 is likely to be lost. In the high-burden countries where progress was already stalling, increased investment and accountability are needed for maternal health care to reverse this trend.

Closing the gap in countries in sub-Saharan Africa and South Asia will require far more than continuing current progress. The annual rate of reduction must be more than doubled in 47 countries to turn the EWENE vision into reality. Whether momentum slows, is maintained or accelerates will mean the difference between life or death for millions of babies. If the trends observed in 2023 continue, mothers in sub-Saharan Africa will endure one in two (50 per cent) of the world's stillbirths between now and 2030, and nearly one in three (30 per cent) of all stillbirths will take place in Southern Asia.

Box 2. Projection scenarios explained

Four projection scenarios provide insight into how current trends in stillbirths could impact future lives and the progress needed to meet critical goals.

Scenario 1: Remaining at 2023 level For each country, the stillbirth rate remains at the 2023 level; in other words, no progress is made from 2024 to 2030.

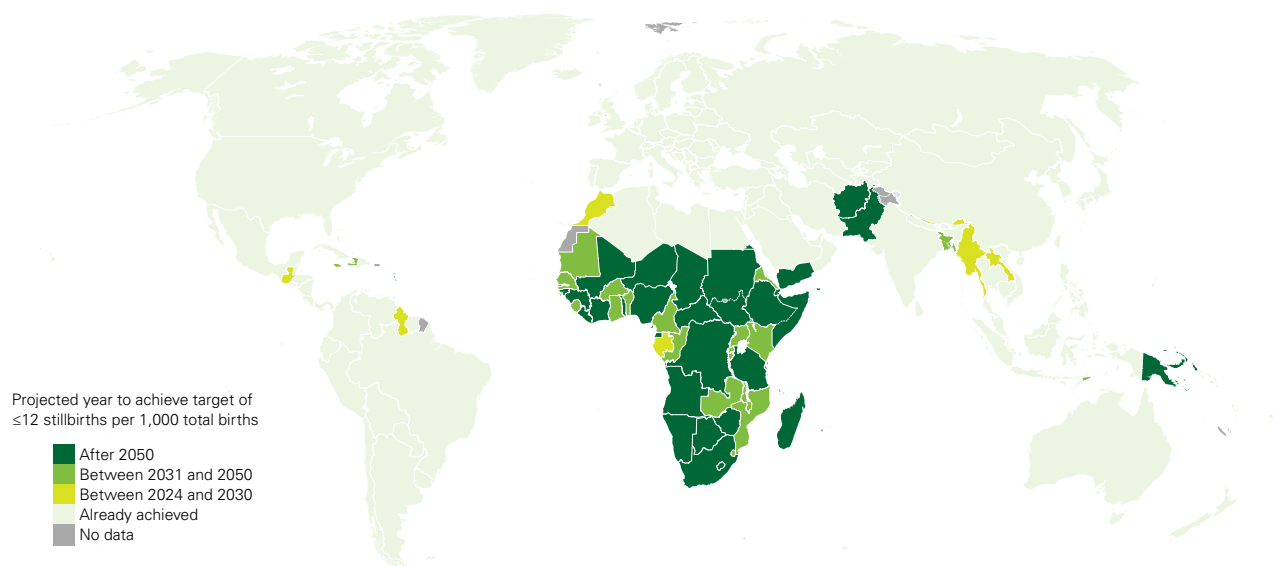
Scenario 2: Continuing current trends For each country, the trend in the stillbirth rate between 2024 and 2030 is equal to the estimated country-specific declines for 2000 to 2023 (as measured by the annual rate of reduction). If a country has an increase in the stillbirth rate from 2000 to 2023, projections from scenario 1 (constant stillbirth rate) are used.

Scenario 3: Achieving EWENE 2030 target For each country, the stillbirth rate in 2030 is 12 deaths per 1,000 total births (the upper bound of the EWENE target), and annual rates of reduction for 2024 to 2030 are calculated based on the country's stillbirth rate in 2023 and the EWENE target; in other words, reductions in the stillbirth rate are accelerated to meet the target by 2030. For countries that have already reached the target or are on track to reach the target before 2030 based on scenario 2, projections from scenario 2 are used.

Scenario 4: Achieving high-income rate The same projection strategy as in scenario 3 is used, except that the 2030 stillbirth rate target is 2.8 stillbirths per 1,000 total births – the average estimated stillbirth rate in high-income countries as of 2023.

About one quarter of countries are off track to meet the EWENE target by 2030

Map 3: Projected year to achieve EWENE stillbirth target if current trends continue, by country



Note: Categories are based on unrounded numbers. Map does not reflect a position by UN IGME agencies on the legal status of any country or territory or the delimitation of any frontiers.

EQUITY IN FOCUS: Disparate risks, uneven burdens

Geography and fragility play critical roles in determining how likely a woman is to have a stillbirth. A recent UN IGME analysis of Demographic and Health Surveys conducted between 2011 and 2023 in 54 low- and middle-income countries (LMICs) (see note under Figure 14) further quantifies these factors, capturing several measurable socio-economic and birth-related factors contributing to the likelihood that a pregnancy ends in stillbirth. These links – which have also been observed in child mortality – demonstrate the need to conduct equity analyses in order to identify vulnerable groups of women within countries when formulating policies and interventions to end preventable stillbirths (see Figure 14).

Household wealth: Women in the poorest wealth quintile experience the highest stillbirth rate (15 stillbirths per 1,000 total births) – or approximately 1.4 times that of mothers in the wealthiest quintile (11 stillbirths per 1,000 total births).

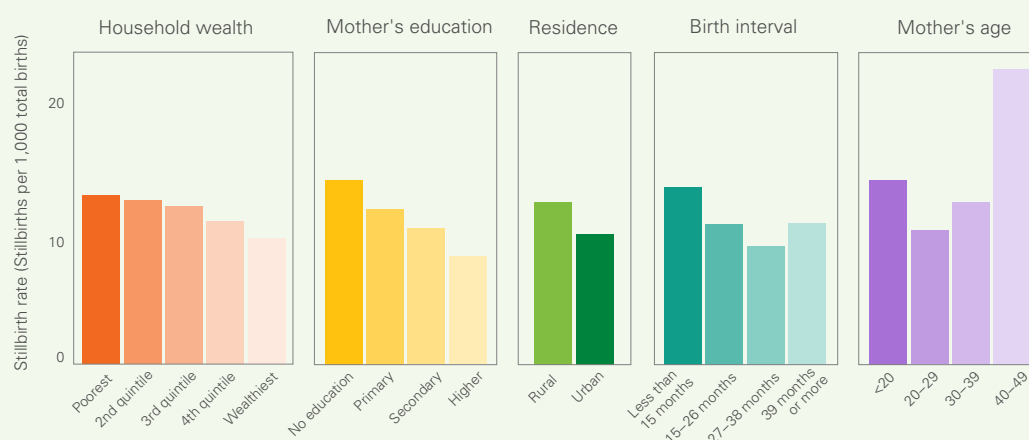
A mother's level of education: The likelihood of a stillbirth among mothers with lower education levels (15 stillbirths per 1,000 total births) is 1.2 higher than that of mothers with secondary or higher education (12 stillbirths per 1,000 total births). As maternal education increases, the risk of stillbirths decreases. Mothers who completed fewer years of school also bear a disproportionately higher burden of stillbirths; while about 53 per cent of live births are to mothers with lower education, they accounted for 60 per cent of stillbirths.

Living in rural vs. urban areas: The risk of a stillbirth is 1.2 times higher in rural areas (14 stillbirths per 1,000 total births) versus urban areas (12 stillbirths per 1,000 total births).

Maternal age and birth interval: Very young or advanced maternal age, as well as short birth intervals, increase the chance of stillbirth. Adolescent mothers have an elevated stillbirth risk compared to women in their twenties or thirties. Older mothers (aged 40–49 years), in particular, face higher stillbirth rates than women in their twenties or thirties, with 26 stillbirths per 1,000 total births versus 13 stillbirths per 1,000 total births, respectively. It is important to note, however, that women of advanced maternal age comprise a relatively small proportion of the births in this analysis, accounting for only 3 per cent of total births, as opposed to adolescent mothers, who comprise 13 per cent of total births. Women who become pregnant within shorter intervals experience a risk that is at least 1.2 times that of pregnancies that occur at longer intervals. The stillbirth rate observed among pregnancies that occurred within 15 months or less of a previous pregnancy was 16 stillbirths per 1,000 total births, compared to a stillbirth rate of 10 stillbirths per 1,000 total births when pregnancies were spaced 27 to 38 months apart.

Risk factors for stillbirths

Figure 14: Stillbirth rates in 54 LMICs, by equity dimensions



Source: The analysis is based on 54 Demographic and Health Surveys (DHS) conducted between 2011 and 2023 in 54 LMICs that accounted for 77 per cent of global stillbirths in 2023. Rates were computed by aggregating stillbirths and births across surveys. After calculating the number of stillbirths and total births in each equity group, the distribution of births by equity group in the surveys was applied to the live births from World Population Prospects 2024 and UN IGME stillbirths for the year 2023 to calculate the aggregated stillbirth rate in each group.

COUNTRIES STANDING UP for stillbirths

Around the world, countries are recognizing the urgent need to include stillbirth prevention and care in policies and programmes. The initiatives highlighted below reflect local- and national-level efforts to reduce stillbirth rates and expand coverage of compassionate bereavement care.

Argentina: Clinical guidelines for fetal and perinatal deaths (2021)

Australia: National Stillbirth Action and Implementation Plan (2020) aim to reduce stillbirth rates by 20 per cent over five years

Brazil: Policy on obstetric racism and reducing maternal and perinatal mortality among minority groups (2024)

Chile: Guidelines for clinical management and social and psychological support of mothers and fathers who have suffered a gestational or perinatal death included in the Domingo Law (2021)

Colombia: Guideline for comprehensive care of gestational and neonatal death and grief in health institutions (2021)

India: National Bereavement Care Package (2024)

Ireland: Health Service Executive National Standards for Bereavement Care following Pregnancy Loss and Perinatal Death (ongoing)

Mexico: Improvements to the health system to expand stillbirth prevention and compassionate care through the Olivia Law (2023)

Nigeria: Project to publish a stillbirth dashboard on the Federal Ministry of Health website, enhance analysis of the District Health Information System 2 stillbirth data and expand the policy and legal environment to reduce stillbirths (2023–2025)

Puerto Rico (USA): Clinical guidelines and protocols in hospitals and health institutions to manage early pregnancy loss and fetal or neonatal death (2016)

Rwanda: Efforts by the ministry-created Rwanda Society of Obstetricians and Gynecologists to expand stillbirth prevention and bereavement services (2025)

Singapore: Parents allowed to register the name of a stillborn child (2024)

South Korea: Extension of leave for female workers after miscarriage and stillbirth (2024)

Sri Lanka: National Bereavement Care Package (2023)

United States: Maternal and Child Health Stillbirth Prevention Act (2024)

United Kingdom: National Bereavement Care Pathway for pregnancy and baby loss (2017); provision of the Pathway across England, Scotland, Northern Ireland and Wales (ongoing)



DATA AVAILABILITY AND DATA GAPS



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Collecting quality data: A team effort

In the health management information system:		
Healthcare workers record, collect and review information on stillbirths as close to the event as possible	Facility delivery data reported to district health office (e.g., as monthly summaries)	District health office reports to national data collection mechanism surveys

Reliable data and systems that track all stillbirths, along with progress made towards related targets and key indicators, are critical to stillbirth prevention. Policymakers need these data to treat and reduce the causes and risk factors linked to stillbirth, and to ensure pregnant women are afforded the same opportunities to see their babies live, regardless of where in the world they are or the personal circumstances of their lives.

Many countries, however, do not regularly collect or produce quality data on stillbirths, with civil registration and vital statistics (CRVS) systems, medical registration systems, HMIS and household surveys failing to record stillbirths. In some cases, stigma or fear of blame stand in the way of accurate stillbirth data collection; in others, equipment, knowledge or technical abilities to report births and fetal deaths may be inadequate.

Where stillbirth data are captured, data quality issues such as the use of non-standard definitions and underreporting or misclassification of stillbirths can render the data unusable. Perinatal death audits are not always carried out following a stillbirth; in both low- and high-income settings, the cause of stillbirth is commonly reported as “unknown” or “unspecified”, leaving the cause of death unknown and families grieving without answers.

Quality data on stillbirths that occur during delivery are also vital. Many health facilities use the appearance of the skin as a surrogate marker for intrapartum stillbirth, but this can be an unreliable measure, and assessment of fetal heartbeat upon labour admission is recommended for all. Data on intrapartum stillbirths must also be reported up to the national level, yet in many contexts, this does not take place.

Data availability for UN IGME's 2023 stillbirth estimates

Data on stillbirths from administrative sources, such as CRVS and HMIS, in addition to household surveys and population studies, inform the UN IGME estimation of stillbirths. Among the 200 countries for which stillbirth estimates are generated, 20 countries have no stillbirth data, and 30 countries lack quality stillbirth data (see Map 4a for available data by data source type in the countries and regions). In 2023, these 50 countries without quality stillbirth data accounted for approximately 23 per cent (400,000) of global stillbirths and 23.2 million live births, or 17 per cent of the world's live births.

Currently, the majority of data (60 per cent) comes from registries (CRVS systems and medical registries), with pronounced differences across income settings. None of the 25 low-income and only 12 of the 51 lower-middle-income countries reported CRVS data for stillbirth; the main data sources for these groups of countries are HMIS (41 per cent), household surveys (21 per cent) and population studies (23 per cent) (see Map 4a). Due to data quality concerns, not all available data can be used to inform the estimates of levels and trends in stillbirths. Nearly two thirds of available data from HMIS (67 per cent), survey (55 per cent)

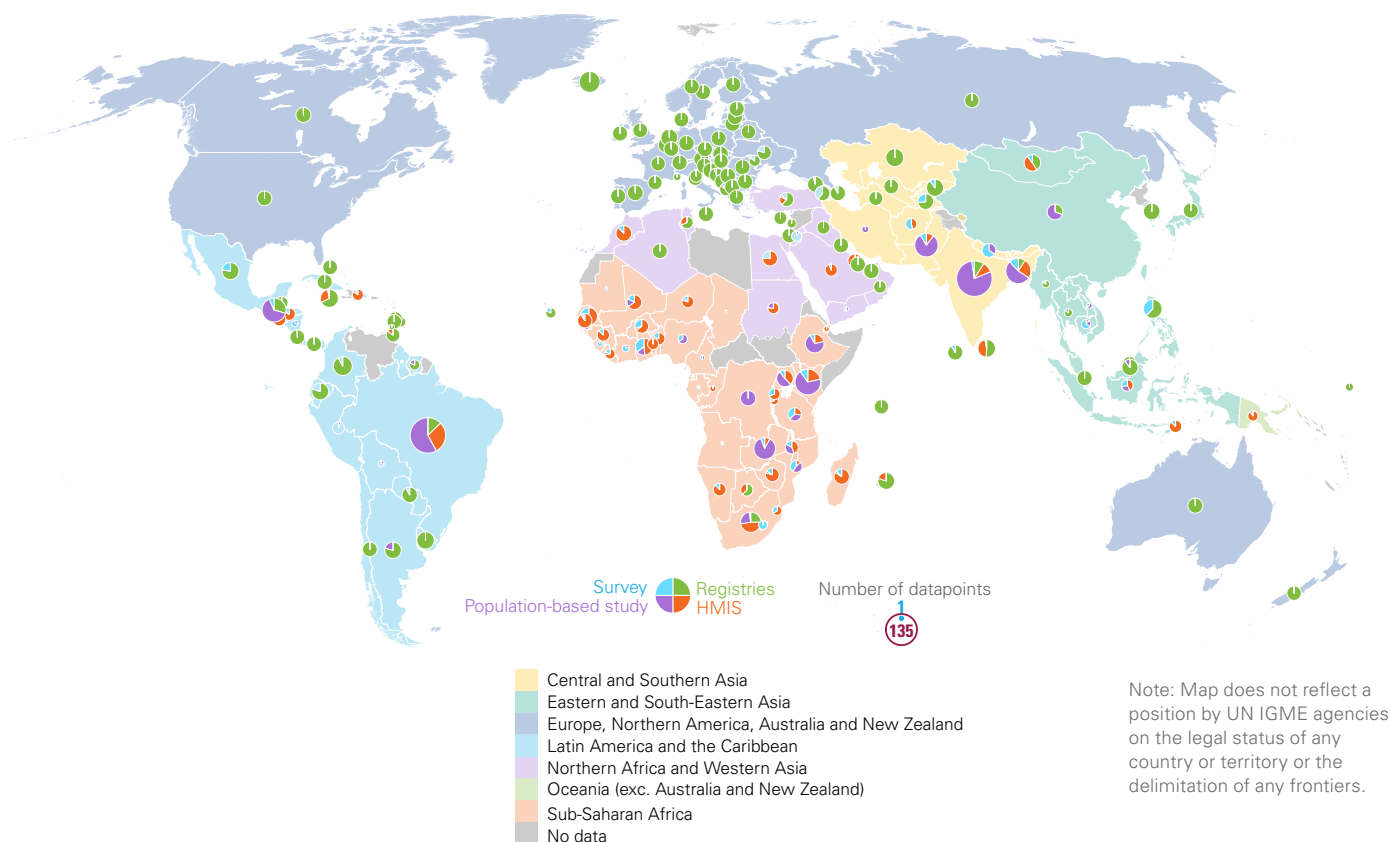
and population study data (68 per cent) – and, to a much lesser extent, CRVS data (29 per cent) – are excluded from the modelling input dataset. In low- and lower-middle-income countries, less than a third of all available data can be used to inform the UN IGME estimates.

Data availability and quality tend to be lowest in the places where stillbirth burdens are the highest. Among the 50 countries lacking quality data to inform the estimates, 32 per cent are in sub-Saharan Africa, and 56 per cent are in the group of low- and lower-middle-income countries (see Figure 15). Conversely, higher data quality is seen in low stillbirth mortality contexts. Only 15 per cent of stillbirth country data produced by high-income countries are excluded in UN IGME modelling due to data quality issues, compared to 70 per cent of data from low- or lower-middle-income countries. As a result, only 16 per cent of the country data informing UN IGME's national stillbirth estimates are from high-burden settings, with estimates for many low- and lower-middle-income countries estimated based on covariates and regional information (see the Annex for details on estimation methodology).

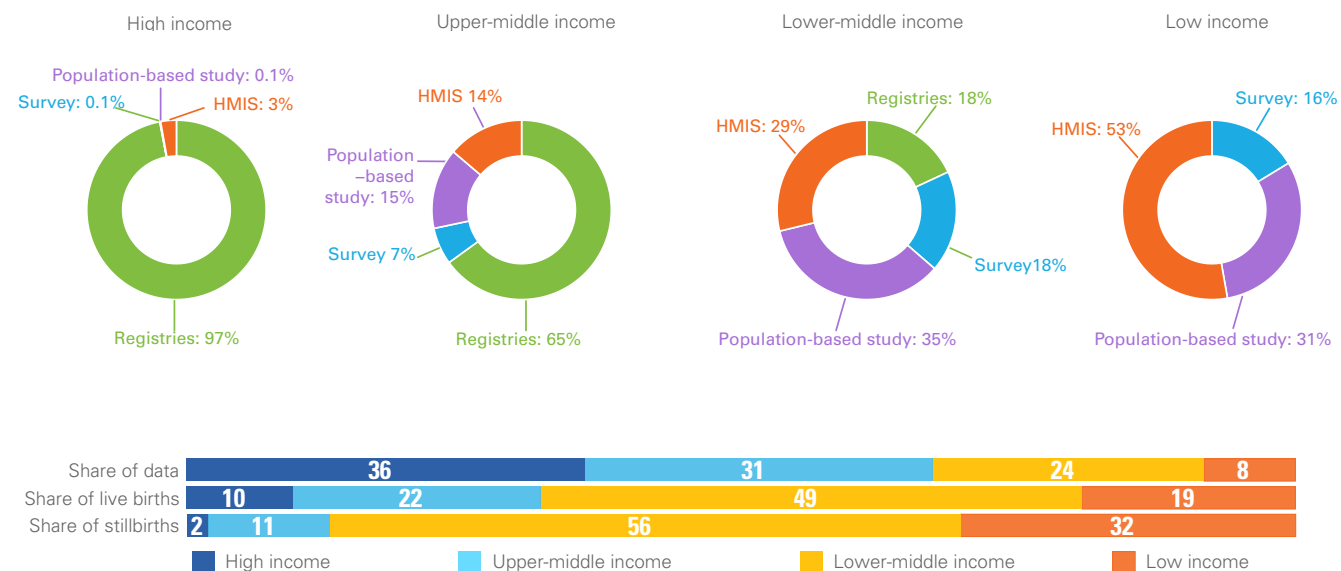


Substantial data gaps – In the absence of vital registration, HMIS and surveys are the main national data sources

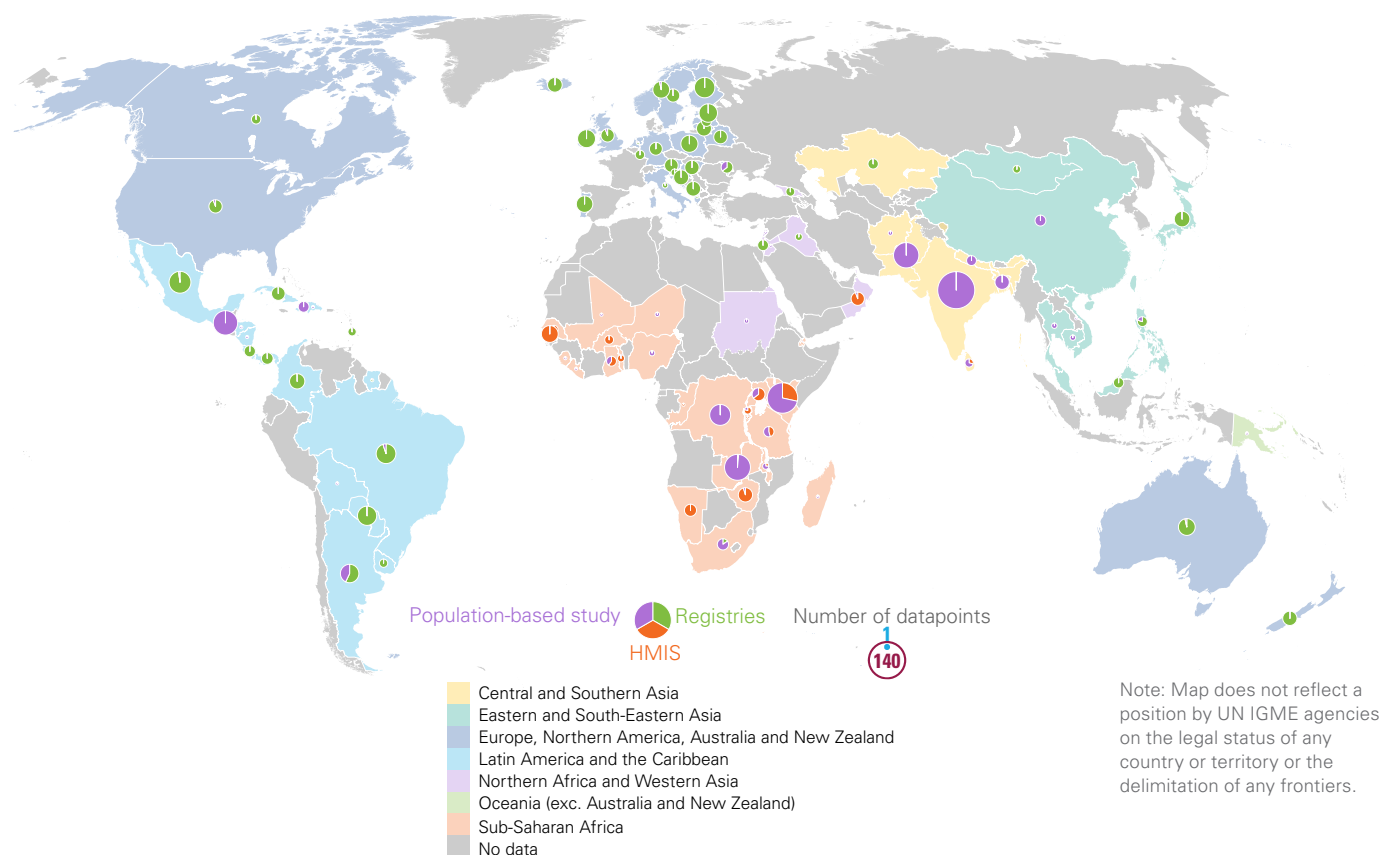
Map 4a: Stillbirth data availability, by country



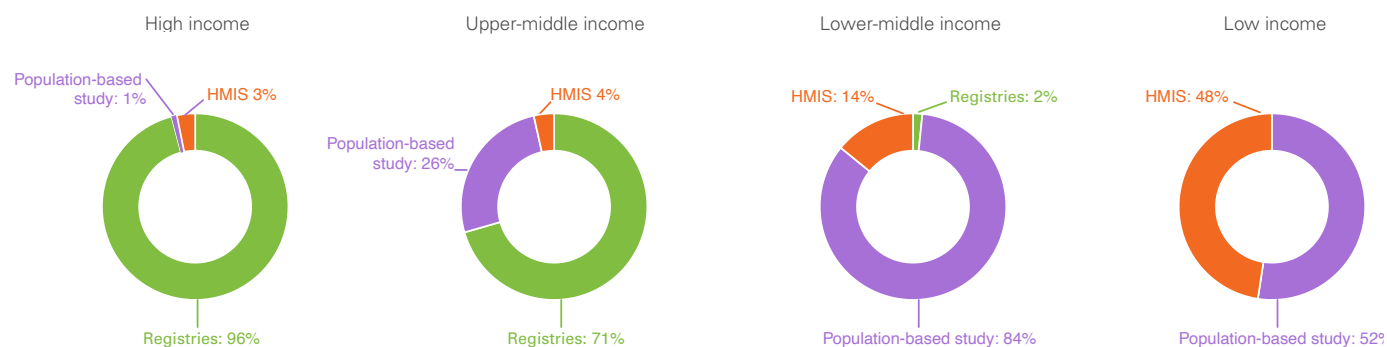
Available data, by source type and income group



Map 4b: Stillbirth by timing data availability, by country



Stillbirth timing data availability, by source type and income group



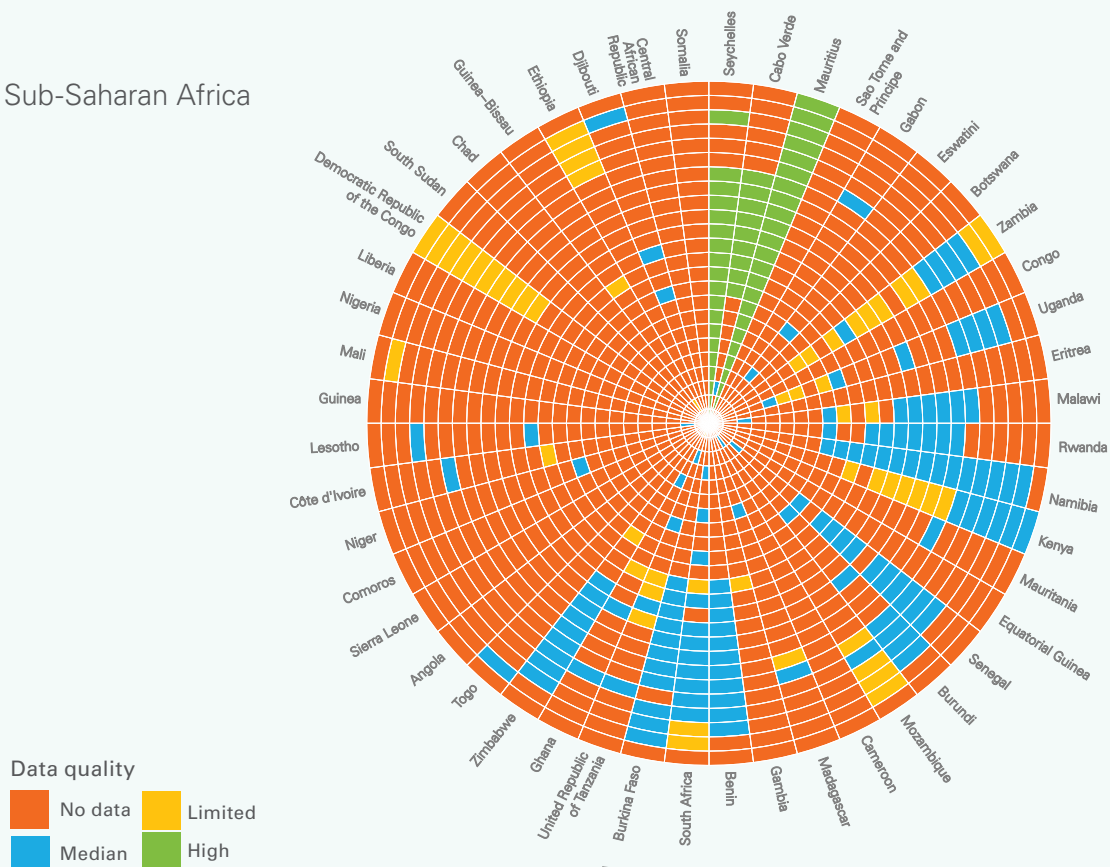
Data on stillbirth by timing are even more scarce. Only slightly less than half (93 countries) report on this crucial piece of information (see Map 4b for available data on timing by country and data source), and for 19 among these the data are of poor data quality. Data are missing across all income groups. In Europe, Northern America, Australia and New Zealand, 55 per cent of countries have stillbirth data by timing, while in sub-Saharan Africa, 48 per cent of countries provide these data.

The assessment method of the timing differs across data sources: 28 per cent use the clinical assessment – fetal heart rate monitoring and ultrasound, 51 per cent rely on fresh versus macerated skin appearance, and the assessment method is unknown for 21 per cent of the data sources. As the most accurate assessment method, the fetal heartbeat should be used when possible to understand the timing of a stillbirth and to start to understand its cause.

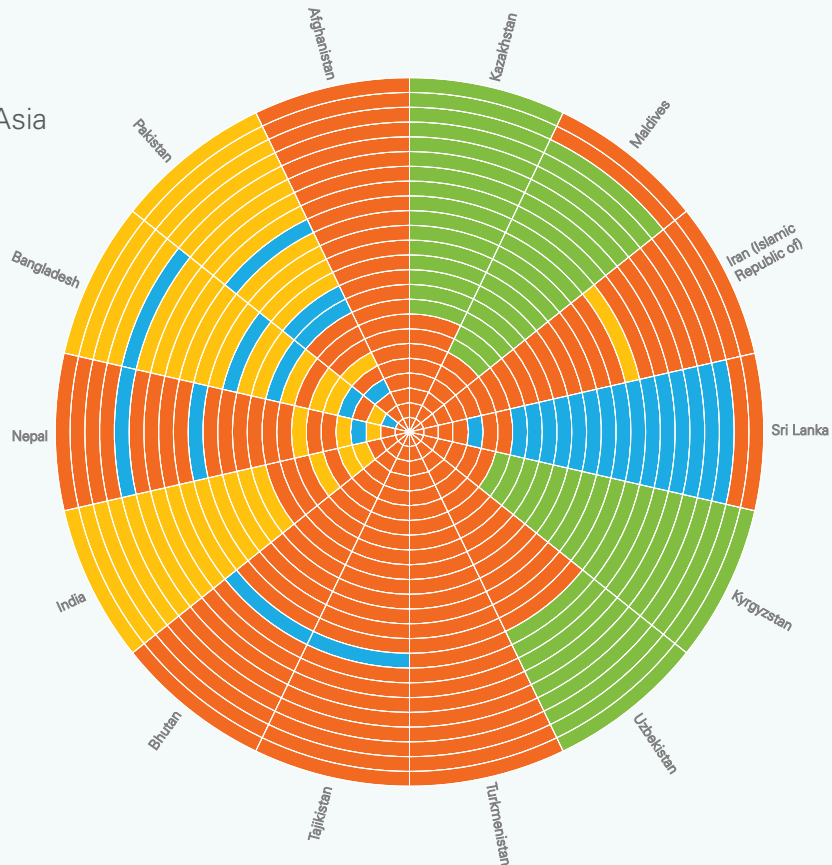
Substantial data quality gaps by data source and region

Figure 15: Data quality by country and by Sustainable Development Goal region (2000–2023)

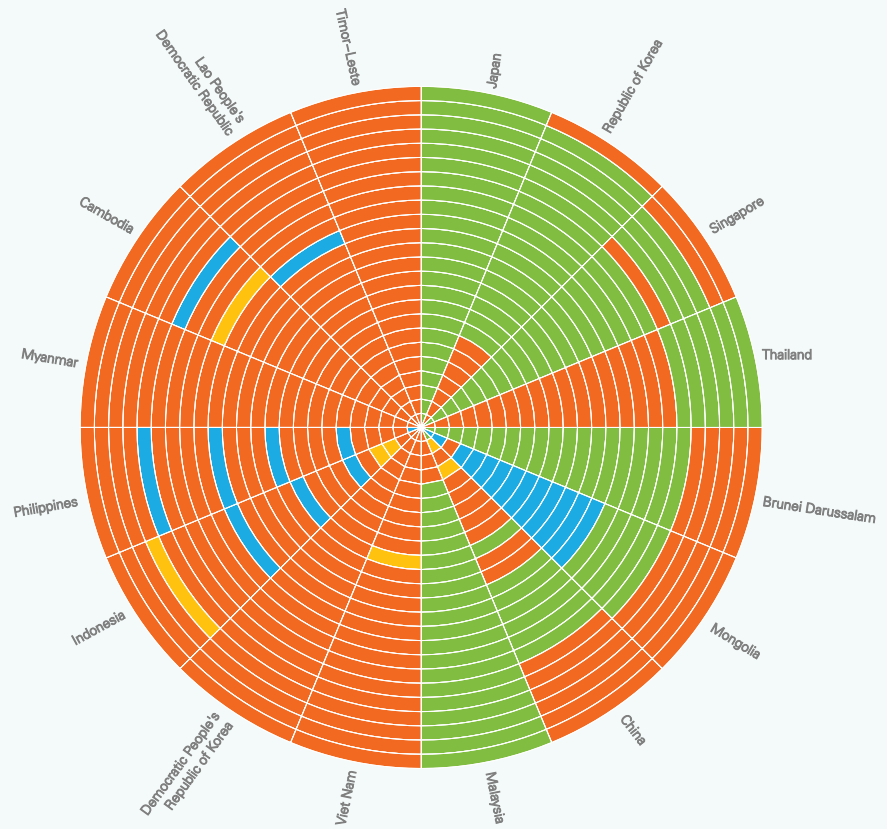
Sub-Saharan Africa



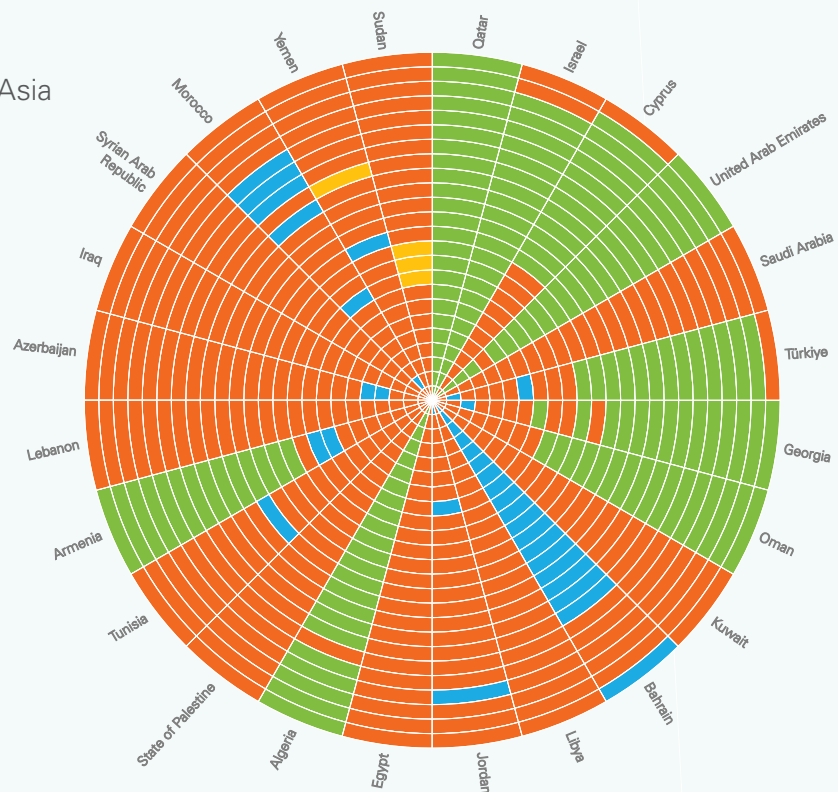
Central and Southern Asia



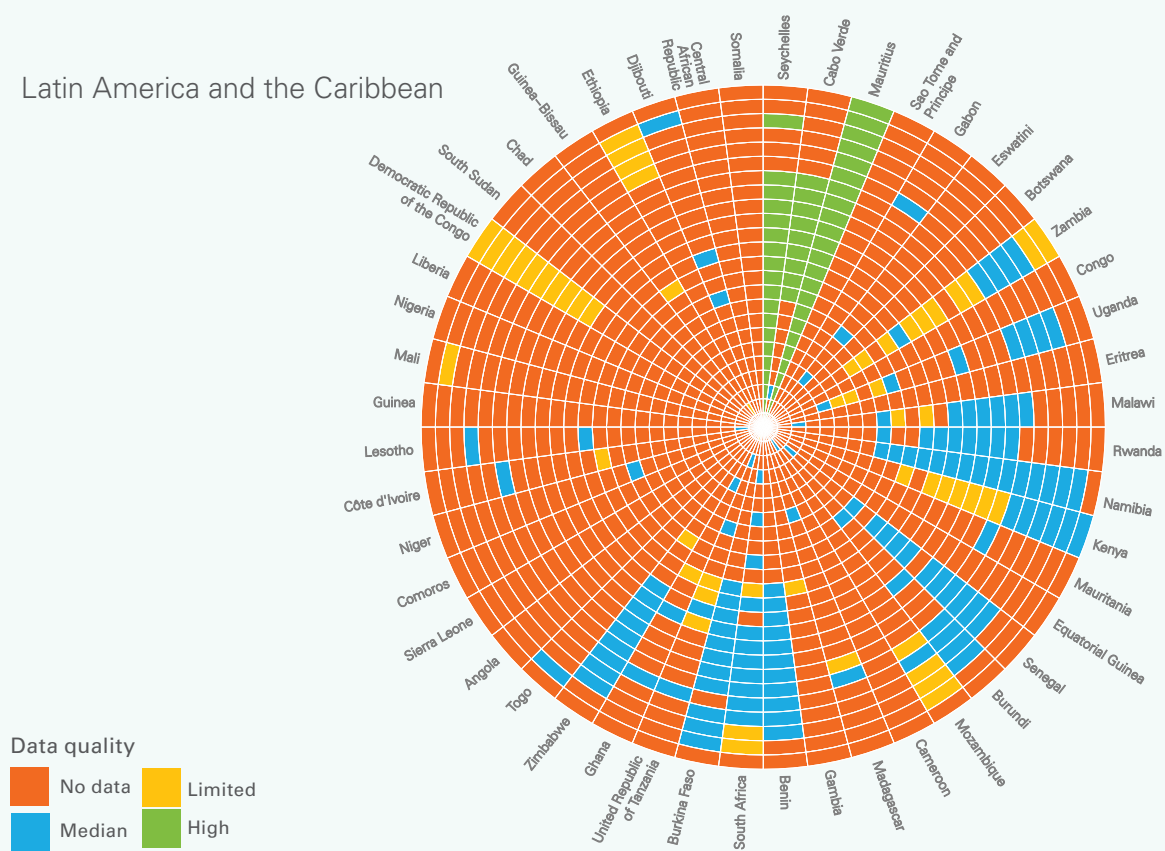
Eastern and South-Eastern Asia



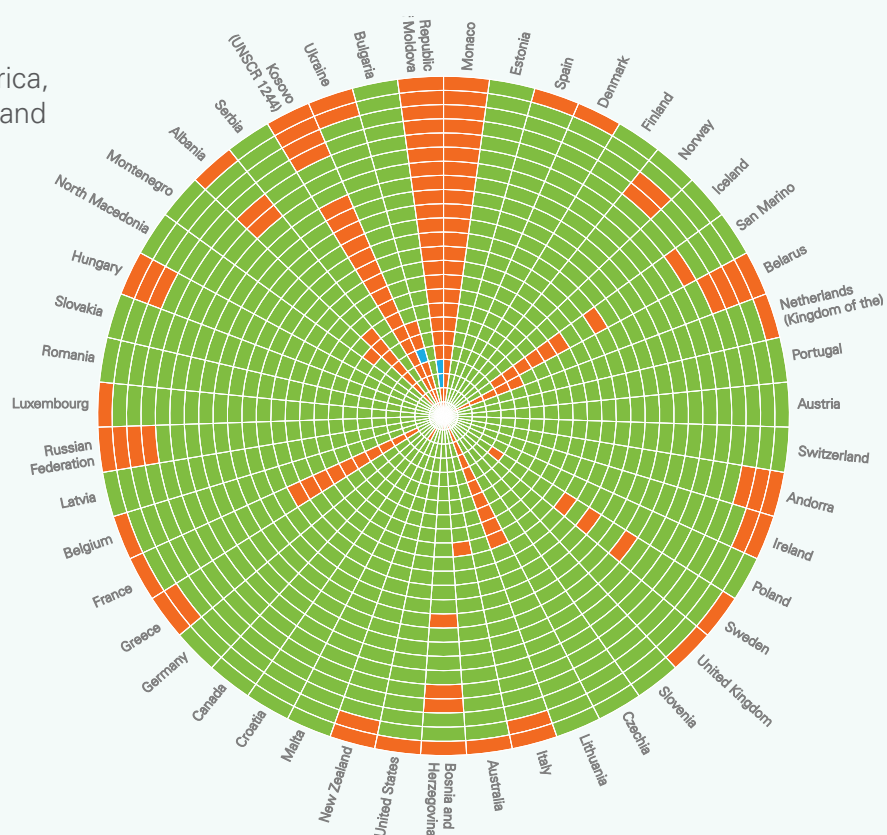
Northern Africa and Western Asia



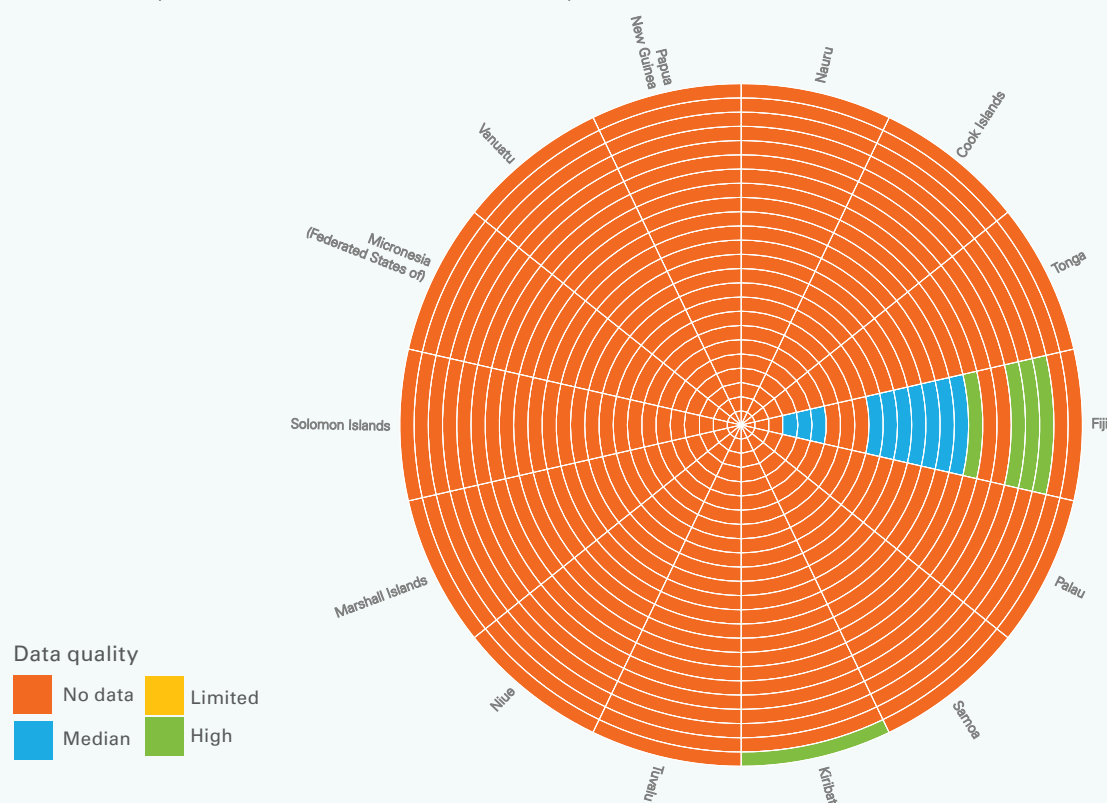
Latin America and the Caribbean



Europe, Northern America, Australia and New Zealand



Oceania (exc. Australia and New Zealand)

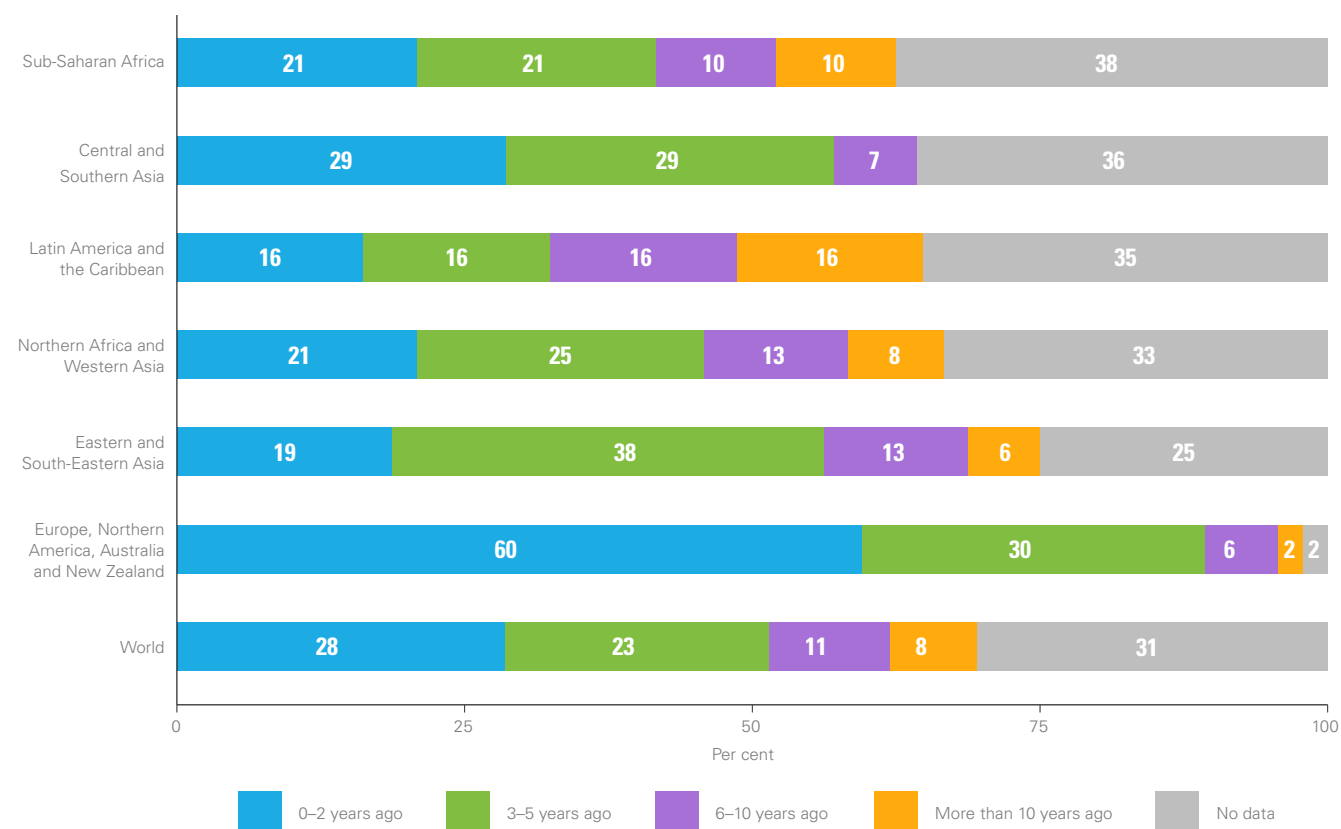


Note: Country-years are classified as having high data quality if registration data are available. Country-years are classified as having medium data quality if data are only available from sources requiring bias adjustments (i.e., surveys or HMIS). Availability of survey data is considered for the period over which the survey was conducted. Country-years are classified as having limited data quality if only population study data are available.

The limited availability of quality data on stillbirths necessitates robust estimation work, as described in this report's Annex. Furthermore, it must be noted that because of these data gaps, there is a large uncertainty around stillbirth estimates. Data systems must be strengthened to improve data availability and quality, particularly in LMICs.

Timeliness of last country datapoint: On average, the most recent data point is more than 3 years old

Figure 16: Distribution of the country extrapolation periods, i.e., the age of the most recent high-quality data point, in the UN IGME 2024 estimation round, by Sustainable Development Goal region



Note: The reference year for a data point refers to the mid-year point of the data collection period.

Among the countries with stillbirth data, on average, the most recent data point is 3.3 years old, with about half of the countries having their latest data point before 2022. For more than 75 per cent of the countries with included data, the most recent data point referred to the last five years (see Figure 16). The uncertainty increases in the estimates for extrapolations past the observation period.

Data availability for late gestational stillbirths

For the estimates in this report, countries were requested to provide stillbirth data using the 28-week-or-more definition. To allow for international comparison, the UN IGME adjusted the stillbirth rate

in cases where data used a different criterion, e.g., birthweight-of-500-g threshold or 22-week threshold. In this round of estimation, 64 per cent of data points used the 28-week-or-more threshold, 25 per cent of data points were adjusted or reclassified to a 28-week-or-more definition, and 10 per cent of data points could not be adjusted because no definition was specified or a nonstandard threshold was used (e.g., 26 or more weeks) – these data points were excluded in the estimation process. The number of countries providing national data (excluding population study data) using a 28-week or more definition has increased from 71 (67 per cent) of 106 countries in 2000–2005 to 108 (83 per cent) of 121 countries in 2020–2023.

Wider burden of loss

WHO ICD-11¹¹ defines stillbirth as birth after fetal death at 22 weeks gestation, and this threshold is recommended for national reporting. However, estimating early gestation stillbirths (those stillborn between 22 and 28 weeks) from national data remains challenging due to variability in the capture of stillbirth data around the thresholds of viability and differences in the recording of late termination of pregnancies across countries, even in high-income settings. For this reason, a 28-week gestational threshold is recommended for international comparison and current UN IGME stillbirth estimates.

However, using the 28-week threshold underestimates the true stillbirth burden. In upper-middle- and high-income countries, estimates from administrative data from registration systems show that the stillbirth rate at 22 weeks or more of gestation is about 1.4 times that of the rate calculated for 28 weeks or more of gestation. In most low- and lower-middle-income countries, where the heaviest burdens of stillbirth are carried, earlier gestation stillbirths are excluded from nationally reported data. Quality study data for this group of countries suggest that the stillbirth rate at 22 weeks or more of gestation is about 1.3 times as high

as the stillbirth rate at 28 weeks or more of gestation in low- and lower-middle-income settings. Across all settings, it is estimated that approximately a third of all stillbirths occur between 22 and 28 weeks, and are excluded from the current round of UN IGME stillbirth estimates. However, variation across country data on the proportion of stillbirths that occur before 28 weeks is large among CRVS, and high-quality study data may be related to data issues rather than true epidemiological differences. To understand the true burden of stillbirths, efforts are needed to increase and improve data collection at earlier gestational timeframes.

To better understand variation in capture at early gestation, the UN IGME requested stillbirth data by detailed gestational age in weeks from week 22 onwards. Forty-five countries, including one lower-middle-income country, the Philippines, provided this detailed data, and Sri Lanka, also a lower-middle-income country, provided detailed weekly data from 28 weeks onwards. Fifty-three countries were also able to provide their stillbirth data by detailed birthweight in 500-g segments. Expanding these data collection and reporting efforts beyond the current level of one fourth of countries is crucial for understanding the real burden of stillbirths.

Stillbirths must be:

- Recognized, recorded, collected and reviewed at the community level and as close to the event as possible
- Reported up from community to district to national levels using standardized definitions and collection mechanisms
- Included in national data collection efforts and in maternal and newborn health data.

Data collection systems such as household surveys and HMIS can inform on levels and trends in stillbirth in the absence of complete stillbirth registration statistics.

- ▶ **Household surveys** that collect information on every pregnancy and its outcome, such as the DHS –are critical to monitoring stillbirths in settings without robust routine data systems. Using pregnancy histories is a crucial step to ensuring that stillbirths and early neonatal deaths are not missed. Surveys can be used to measure the stillbirth rate retrospectively, identify determinants of stillbirth and assess inequalities.
- ▶ **HMIS** can be used to collect detailed data by gestational age, birthweight, and report data. To provide health care workers and data managers further guidance, the UN IGME has published a guideline¹² that addresses both definitions, criteria for collecting information when stillbirths occur, and

the assessment of data quality and production of statistics in HMIS.

- ▶ **Registration systems** *CRVS systems*: Stillbirths are a vital event and should be recorded by detailed gestational age and birthweight to produce timely statistics for national and international comparison. Improving the completeness of stillbirth registration needs to be included within CRVS strengthening efforts. Complete stillbirth data from registration systems allow timely data and can be disaggregated at the subnational level where stillbirths occur. The health system can play an important role in reporting stillbirths to the CRVS.¹³
- ▶ **Medical registries**: In several countries, stillbirths are recorded and reported by the health sector to the Ministry of Health.



Box 3. In policies and monitoring, a neglected issue

Data from 106 countries in the ENAP-EPMM progress tracking tool in 2023 (now EWENE)

How many countries have set targets to reduce...

- ▶ the stillbirth rate **29%**
- ▶ maternal deaths **83%**
- ▶ neonatal deaths **83%**

How many countries have critical maternal and newborn death surveillance and response systems in place?

- ▶ **57%** have one for maternal and perinatal deaths
- ▶ **76%** have one for neonatal deaths
- ▶ **89%** have one for maternal deaths

What policies are in place to ensure every birth and death is registered and recorded?

As reported by 106 countries in the Every Woman, Every Newborn, Everywhere (EWENE) tracking tool:



98%

have a mechanism for birth registration



83%

have a mechanism for neonatal death registration



61%

have a mechanism for stillbirth registration



7 THINGS TO DO: TAKING A STAND FOR STILLBIRTH

7 THINGS TO DO: Taking a stand for stillbirth

1) Improve the knowledge and evidence base through better measurement of stillbirths, including working to:

- Standardize and align stillbirth definitions and measures with international standards
- Capture stillbirths on data platforms and audits – e.g., routine HMIS, registers and monthly reporting forms, and perinatal death surveillance and response systems
 - Increase investments in surveys that lead to more detailed pregnancy histories
- Provide training and support to health care workers, data collectors and programme managers to facilitate registration of stillbirths
- Collect information on timing and pathways of stillbirth
- Expand data ownership and use, especially at the facility and district levels
- Report stillbirth data and review them alongside data on maternal and neonatal deaths
- Have stillbirth data travel up the chain to the national level, where progress and the most at-risk women can be identified and monitored

2) Invest in health care systems to ensure high-quality care, including investments in:

- Comprehensive and empowering family planning services, education and programmes available to all women
- Universal health coverage offered along the continuum of care
- Quality, respectful care and transportation and referral pathways are provided antenatally, during labour and after death
- The health care workforce – especially midwives, nurses and community health workers – and facilities themselves so that they are well equipped to deliver quality care during pregnancy and labour
- Recognition that the risk factors and causes of stillbirths are inseparable from those tied to maternal and neonatal mortality and measures known to reduce stillbirths, and integration of this information into routine Reproductive, Maternal, Newborn, Child and Adolescent Health care, such as:
 - Periconceptional folic acid supplementation or fortification
 - Insecticide-treated bed nets and intermittent preventive treatment with antimalarial drugs
 - Syphilis and infections screening, detection and early treatment
 - Detection and management of chronic diseases such as hypertensive disorders of pregnancy, diabetes and obesity
 - Detection and management of fetal growth restriction and identification
 - Birth preparedness plans and skilled birth attendance through trained midwives
 - Access to emergency obstetric care
- Cooperation and communication between health care providers and partners working with pregnant women to improve the health of the woman-baby dyad throughout pregnancy and childbirth
- Incorporation of WHO's recommendations on antenatal and intrapartum care for a positive pregnancy experience

3) Establish stillbirths as a national health priority and set EWENE-aligned local and national stillbirth targets, including:

- Clear goals and political will to accelerate progress towards reaching targets on child survival and health
- Commitments that match those made to reducing neonatal mortality
- Local and national development plans, policies and investments designed to realistically achieve goals

4) Reduce stigma and promote stillbirth prevention by making:

- Data stronger and more widely available, providing crucial evidence on which women face the highest risks and where prevention interventions are most needed
- Every stillbirth examined as part of perinatal death review with response plans
- Stillbirth included in all relevant maternal and newborn health reports, policies, investments and research programmes
- Awareness-raising campaigns speak candidly about stillbirth in communities and address its long-lasting impacts; tackle social taboos, stigma and misconceptions; and widely communicate that most stillbirths are preventable
- Safe spaces for health care workers and expecting and bereaved parents to freely speak about stillbirth, including the right to compassionate care, participate in their child's death review, and grieve
- The significant direct, indirect and intangible costs of stillbirth recognized by communities, health care settings and economies

5) Provide compassionate and respectful care, including ensuring that:

- Every stillbirth is recognized as an individual and unforgettable story of loss and is investigated in order to identify contributing factors
- Bereaved mothers receive explanations of why their child has died
- Health care workers are equipped to sensitively address stillbirth and support families through the grieving process
- High-quality bereavement support and services are available to mothers, families and health care workers, including information and support for future pregnancies

6) Identify and address inequities, including efforts to:

- Accelerate prevention of stillbirths in sub-Saharan Africa in anticipation of the expected increase in the number of pregnancies and total births in the region
- Renew and refocus attention to Southern Asia's high-burden countries
- Analyse global stillbirth prevention and care in fragile- and conflict-affected settings
- Examine and measure stillbirth risks among vulnerable and marginalized populations in every country and use the data to drive action to close these gaps

7) Sustain financing levels by:

- Earmarking funding, support and human resources in national and subnational plans to ensure every pregnant woman and newborn has access to high-impact interventions like antenatal care, skilled attendance at birth and emergency obstetric care
- Continuing and expanding global political commitments and investments to end preventable stillbirths

It is time for the taboos and inaction that surround stillbirths to end. Prevention of stillbirths and care for bereaved women and their families must be included and prioritized in data collection efforts, policies and programmes and become a routine part of the Reproductive, Maternal, Newborn, Child and Adolescent Health continuum of care. Improving access to timely quality of care throughout pregnancy and around the time of birth for every woman everywhere will yield a quadruple return on investment – reducing stillbirths, improving maternal and newborn health, and advancing child development.

Global progress for every mother and baby begins with local action. Clinicians, nurses, midwives and community health workers must be supported to better understand why stillbirths occur and which women face the highest risks – and then take the needed measures to prevent them.

Bereaved women and their stillborn babies must be treated with dignity and respect. When a stillbirth occurs, mothers deserve compassion and care, not blame or shame. Babies who are stillborn have a right to be counted and mourned, and their parents have a right to see their child's death properly documented and honoured.

Stronger commitments – including sustained funding – are urgently needed to end preventable stillbirths in every country around the world. UNICEF, WHO and partners call for efforts to address these preventable deaths and end their neglect in policies and programmes. Standing up for stillbirths is directly tied to every country's upholding of the rights of mothers and children to quality and timely antenatal and delivery care.





ANNEX

Estimating stillbirth rates

This chapter summarizes the methods the UN IGME uses to generate stillbirth estimates and the proportion of stillbirths by timing. Since 2018, the UN IGME has been working on estimating stillbirth indicators.

Strategy

The UN IGME's approach to estimating stillbirth rates (SBRs) includes the following steps:

1. Compile all available stillbirth data at a country level, derived from administrative sources, household surveys or population-based studies.
2. Evaluate data in accordance with the data quality criteria and produce adjustments or recalculations by applying standardized definitions.
3. Estimate global and country-specific trends of SBRs using a smoothing time series model,

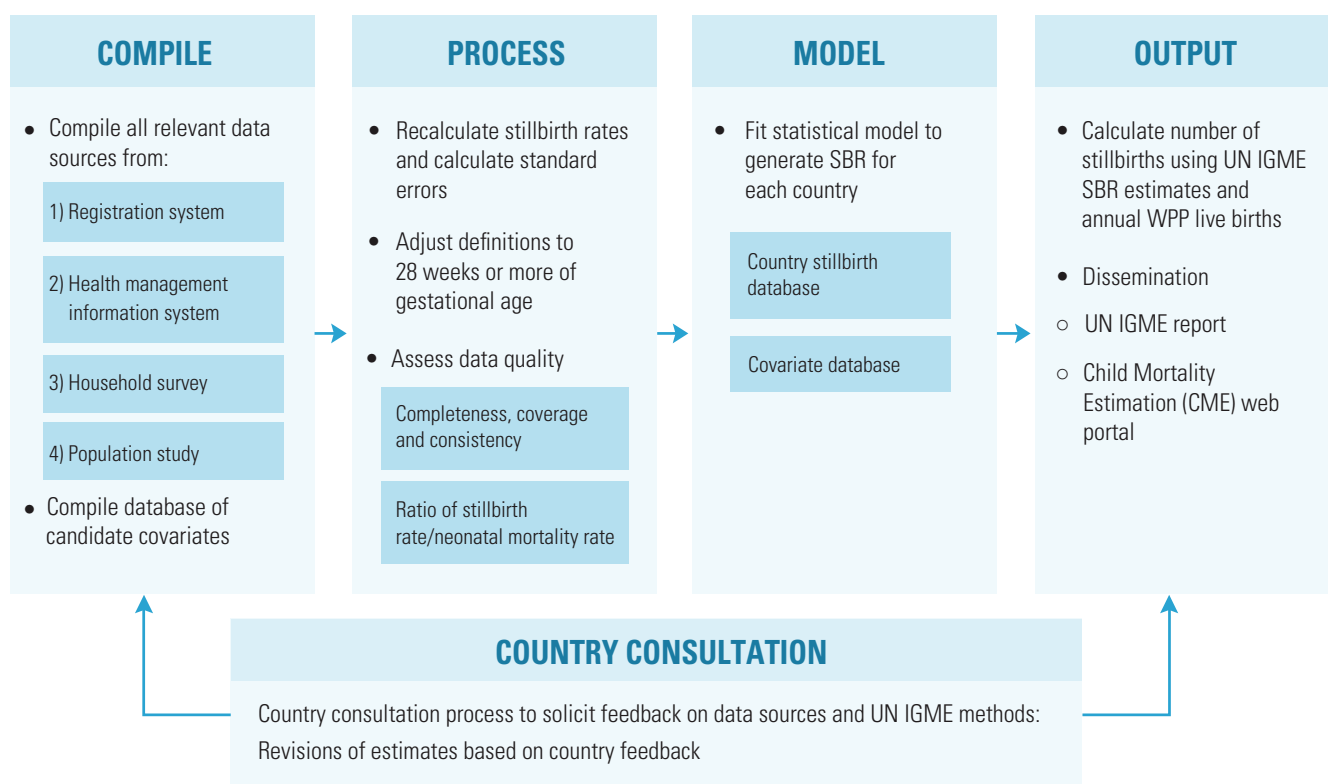
supplemented with covariates associated with SBRs. This process averages empirical data on stillbirths derived from the different sources available for a given country. In the case of countries with sparse or no data, the identified covariates associated with stillbirth will inform the trends in SBRs.

Estimates by the UN IGME may differ from the official statistics by Member States, which may use alternative but equally rigorous methods. To increase the transparency of the estimation methodology and make stillbirth data available to users worldwide, the UN IGME makes all data sources and stillbirth estimates available on its web portal at <http://www.childmortality.org>.

Stillbirth concept and definition

In the UN IGME estimation work, consistent with the International Classification of Diseases (ICD),^{14,15} only 'late gestation fetal deaths' are included in UN IGME

Figure 17: The UN IGME approach



*Throughout the document, 'stillbirth' refers to third trimester (late) stillbirth.

international stillbirth monitoring. The SBR is defined as the number of babies born with no sign of life at 28 weeks or more of gestation, per 1,000 total births.

The stillbirth rate is calculated as:

$$SBR_t = 1000 * \frac{sb_t}{sb_t + lb_t}$$

sb_t refers to the number of stillbirths stillbirths ≥ 28 weeks of gestational age

lb_t refers to the number of live births regardless of gestational age or birthweight

SBRs using gestational age are not equivalent to those using birthweight criteria. Therefore, to improve comparability of stillbirth data from different countries UN IGME uses a stillbirth definition that has the gestational age as a single criterion in line with the ICD-114, 5 recommendations for standardized recording and reporting of perinatal deaths. Using these definitions also improves data accuracy and international comparison. Gestational age is preferred over birthweight and length criteria as it is a better predictor of maturity and hence viability and is the most commonly used criteria across data sources including household surveys.

Key terms

Birthweight: The first weight of the fetus or neonate obtained after birth. For live births, birthweight should preferably be measured within the first hour of life before significant postnatal weight loss has occurred.

Gestational age: The duration of pregnancy estimated based on the best obstetric estimate of gestation. Gestational age is counted by calendar days where day 0 is used to refer to the first calendar day of gestation and day 1 for the second calendar day.

Live birth: A live birth is the delivery of a baby, irrespective of the duration of the pregnancy, which after such separation shows signs of life. Signs of life at birth include breathing, beating of the heart, pulsation of the umbilical cord and definite movement of voluntary muscles whether the umbilical cord has been cut or the placenta is attached. Fleeting reflex activity, defined as automatic involuntary reflexes triggered by stimuli such as touch or temperature changes, observed only in the first minute after birth does not warrant classification as a sign of life.

Stillbirth: The delivery, or birth, of a baby after a **fetal death** has occurred. Delivery typically occurs spontaneously or planned in consultation with health providers and affected parents, within days of fetal death. Occasionally, weeks or longer pass between the fetus' death and its delivery or stillbirth.

A stillbirth is a baby born following a fetal death at 154 days (22+0 completed weeks) or more of gestation.

Gestational age subgroups

Early gestation stillbirth is a stillbirth between 154 and 195 days of gestation (22⁺⁰–27⁺⁶ weeks).

Late gestation stillbirth is a stillbirth at 196 or more days gestation ($\geq 28^{+0}$ weeks).

If no gestational age information is available birthweight ≥ 500 g can be used as a proxy for stillbirth, with early gestation stillbirths ≥ 500 to $<1,000$ g and late gestation stillbirths $\geq 1,000$ g.

Stillbirths by timing

Antepartum stillbirth is a stillbirth following antepartum fetal death (before onset of labour). Intrapartum stillbirth is a stillbirth following intrapartum fetal death (occurring during labour).

Every effort should be made to assess and record the vital status of the fetus for all women presenting in labour to the health facility. Skin appearance is a poor proxy for stillbirth timing and should only be used when the vital status of the baby at the onset of labour, or admission to the health facility is unknown. In these cases, macerated stillbirth (presence of maceration at delivery) suggests antepartum death, while fresh stillbirth (no maceration) suggests intrapartum death.

Total births: The sum of stillbirths and live births. If the lower limit of stillbirth used differs from 154 days (22+0 completed weeks), for example ≥ 196 days ($\geq 28^{+0}$ weeks) for international reporting, this should be clearly stated.

Data sources

Estimates of SBRs for a country can be derived from various sources, such as administrative data (e.g., vital registration systems, birth or death registries, or health management information systems [HMISs]), household surveys or population-based studies obtained from a review of academic literature.

Data from registration systems are the preferred data source for estimating stillbirths by the UN IGME. The reliability of stillbirth estimates depends on the accuracy and completeness of reporting and recording of births and deaths. Not all countries maintain a timely and complete registration system for stillbirths. As a result, stillbirth data from registries can be biased due to underreporting or misclassifications. Moreover, in many low- and middle-income countries (LMICs), stillbirths are not reported in registration systems at all.

Household surveys – such as the Demographic and Health Surveys (DHS) supported by the United States Agency for International Development (USAID),¹⁶ the UNICEF-supported Multiple Indicator Cluster Surveys (MICS)¹⁷ and other nationally representative surveys – are another source of data on stillbirths in LMICs. In addition, in several LMICs, data from population-based studies provide an important data source on stillbirths.

Data on stillbirths are systematically collected and compiled by the UN IGME: The current database used to estimate SBRs is available on the UN IGME web portal (<<http://childmortality.org/>>). In total, the database contains almost 3,500 country-year data points from 175 countries from the year 2000 to 2023. Most of these data points (2,000, or 58 per cent) are from administrative systems, including civil registration and vital statistics (CRVS) systems and medical birth and death registries. Of the remaining data points, 608 (18 per cent) are from HMIS systems, 287 (8 per cent) are from household surveys, and 570 (16 per cent) are from population studies. After assessing the data quality, about 1,900 data points (55 per cent) were included in the estimation model.

Figure 17: Data sources for stillbirth data in countries

1. Administrative data

Collected data from registration systems and health data systems, including data from HMISs

2. Household survey data

Collected data through pregnancy histories or RCs

3. Population study data in LMIC

Literature review of population studies

► Administrative data

The majority of administrative data come from registration systems and health data systems, including HMISs (see data sources 1 in Figure 17). Often, data from registration systems record stillbirths and live births using detailed gestational age or birthweight. HMIS data are collected in health facilities, and in many countries, the District Health Information System-2 (DHIS2) is the most common HMIS data platform. Few HMIS systems currently report detailed gestational age or birthweight data on stillbirths.

► Household survey data

Information on stillbirths in household surveys can be collected in two different ways: with a full pregnancy history (PH) or with a reproductive calendar (RC) (see data source 2 in Figure 17).^{19,20,21} In the PH, women of reproductive age are asked about all pregnancies in their lifetime. For each pregnancy, they are asked to provide information on the duration of the pregnancy, the outcome of the pregnancy (e.g., miscarriage, stillbirth or live birth) and the date of birth or end of

pregnancy. In the RC, women are asked about the duration and month of pregnancy end for pregnancies that did not end in a live birth in the last 60 months. RCs are usually administered alongside a full birth history that collects data from women of reproductive age on every live birth.

In pregnancy histories, the SBR is calculated as the number of stillbirths that occurred in the seventh month or later, divided by the total number of stillbirths and live births. In some surveys with PH modules, the women were only asked whether they had a stillbirth and the date of the stillbirth. In these cases, a seven-month duration of pregnancy was assumed. In some survey-specific cases, a stillbirth was defined by the questionnaire as a fetal death occurring in the fifth or sixth month or later. In RCs, the SBR is the number of pregnancies that are terminated in the seventh month or later of pregnancy divided by the number of pregnancies that reached at least the seventh month. PH data allow the calculation of SBRs for specific time periods in the past. Where the microdata were available, UN IGME recalculated the stillbirth estimates with standard errors from PH and RC. For PH data, the stillbirth estimates were calculated for 5-year calendar periods and for five intervals (e.g., 25 years) before the survey date. The most recent 5-year calendar period was included in the estimation model. The RC data allow the calculation of SBRs for the 5-year period preceding the survey. However, stillbirth estimates from the RC were not included in the model if estimates from the PH in the same survey were available.

► Population studies on stillbirth

Another source for data on stillbirths is subnational population-based studies (see data source 3 in Figure 17). Subnational population-based study data were sought for all countries without high coverage of routine administrative data. The literature review undertaken for the previous stillbirth estimates²² was updated through 30 June 2024. In addition, further reanalysed population-based stillbirth data

were obtained from a UNICEF data call to maternal-newborn health experts.'

► Comparability of stillbirth data across data sources

The lack of a standard application of definitions for stillbirth in many data sources results in comparability challenges for the assessment of SBRs between settings and over time. Stillbirths are reported by different gestational age week cut-offs, ranging from 16 weeks to 28 weeks or more, or by birthweight ranging from 500 g to 1,000 g or more, or by a combination of gestational age and birthweight. In several cases, the data source provides no clear definition. Using different gestational age and birthweight thresholds will inevitably produce different estimates of the SBR. Missing information on gestational age or birthweight can also impact overall data comparability, especially when the proportion missing such information is large. As a result, the UN IGME adjusts stillbirth data to the 28 weeks of gestation or more definition, taking into account missing gestational age or birthweight data where possible (see section 4.2).

► Covariates

To inform SBRs in the case of countries with little or no data, the estimation model included factors associated with SBRs as covariates. The candidate covariates were based on a conceptual framework identified from published literature in 2016 by Blencowe et al.^{23,24} The framework includes distal determinants such as socio-economic factors, interacting and overlapping demographic and biomedical factors, associated perinatal outcome markers and markers for access to health care. The covariate data are smoothed with a time series trend to reduce small fluctuations in measured covariates. The covariates that had the most explanatory power were selected for inclusion in the model. Table A1 lists the selected covariates for the estimation model.

Table A1. Selected covariates indicators and data sources

Indicator	Data source
Antenatal care 1+ visits: Percentage of women (age 15–49) who were attended to at least once during pregnancy by skilled health personnel.	World Health Organization (WHO)/ UNICEF: DHS, MICS and other national household surveys
Protection at Birth (PAB) against tetanus with Tetanus Toxoid (TT): Percentage of pregnant women protected by tetanus toxoid-containing vaccines (TTCV) who would give birth to a child protected against tetanus as a result of maternal transfer of antibodies through the placenta.	WHO/UNICEF: Modelled based on administrative reporting; TT coverage surveys
C-section rate: Percentage of deliveries by Caesarean section.	UNICEF: DHS, MICS and other national household surveys
Gross national income (GNI) per capita, purchasing power parity (PPP) (constant 2021 international dollar): GNI per capita based on PPP. PPP GNI is GNI converted to international dollars using PPP rates. An international dollar has the same purchasing power over GNI as a US dollar has in the United States. GNI is the sum of value added by all resident producers plus any product taxes (less subsidies) not included in the valuation of output plus net receipts of primary income (compensation of employees and property income) from abroad. Data are in constant 2021 international dollars.	World Bank, International Comparison Program database
Low birthweight: Percentage of live births that weighed less than 2,500 g (less than 5.51 lb).	UNICEF/WHO Low Birthweight Estimates, 2023 edition
Neonatal mortality rate (NMR): Probability of dying in the first 28 days of life, expressed per 1,000 live births.	UN IGME: Modelled based on data from vital registration, household survey and population census
Annual percentage of population at mid-year residing in urban areas. Urban population refers to people living in urban areas as defined by national statistical offices.	UN Population Division, World Urbanization Prospects 2018

Methodology to estimate SBRs

To reconcile differences across data sources and better account for the systematic biases associated with the various types of data inputs, members of the Core Stillbirth Estimation Group of the UN IGME have developed a new approach to make decisions regarding data exclusion, analyse the definitional adjustments needed and fit a smoothed trend curve to a set of observations that are described below. The estimated trends are extrapolated to provide estimates through to 2023. A more detailed technical description can be found in Wang et al.²⁵

► Data quality assessment

The UN IGME assessed the quality of the stillbirth data from the four types of data sources used to evaluate completeness and consistency. Data

were excluded if: they lacked a clear source of definition or clear information on data collection systems; a high proportion of reported stillbirths had unknown gestational age or birthweight; data were internally inconsistent; or coverage of live births in administrative data systems was estimated below 80 per cent. Vital registration data with incomplete coverage of child deaths were also excluded. Consistency across data sources was further assessed by comparing stillbirth estimates to similar data sources within the same country and expected global and regional patterns in mortality.

As part of the assessment of data quality, the plausibility of the ratio of SBRs (as per the 28 weeks of gestation or more definition) to neonatal mortality rates (NMRs) was assessed by comparing these ratios to the distribution of ratios obtained from high-quality

LMIC study data. High-quality LMIC study data are defined as population-based prospectively-collected data with recruitment prior to 28 weeks of gestation, and follow-up to at least 28 days of age of live births.

In assessing the NMR to SBR ratio in the input database, the NMR from the data source was used where available. Where data sources had missing NMR data, the estimated NMR by UN IGME²⁶ was used. For observations from HMIS and population studies on stillbirths, the ratio of observed SBR to the UN IGME NMR was calculated, and the same exclusion approach was applied so that observations with extremely low SBR compared to national-level NMR were excluded. In summary, the mean and variance of the setting-specific SBR:NMR ratios were estimated, assuming that each observed SBR:NMR ratio was the sum of a setting-specific SBR:NMR ratio and random stochastic error.

If stillbirths were under-reported relative to neonatal deaths for a specific observation, the associated observed ratio of SBR to NMR would be lower than the true ratio. To quantify whether an observed ratio was 'extremely' low, the probability of observing a ratio that is smaller than the observed ratio was calculated (taking account of the uncertainty associated with the observed ratio) using the distribution of ratios obtained from the high-quality data. If this probability was less than 0.05, the observation was excluded from the database. This approach was applied to all observations in the database with 28 weeks of gestation or more definitions and adjusted 28-week definitions.

► Definitional adjustment of stillbirth data

SBR estimates were constructed based on using a stillbirth at 28-week gestation or more definition. If information based on the 28-week definition was not available, observations recorded using alternative definitions were adjusted as described below prior to being used in the model fitting. Bias and additional uncertainty associated with alternative definitions were taken into account in the model fitting for such observations.

For LMICs, high-quality data from LMIC studies were used to calculate the conversion, while for high-income countries, national administrative data were used. For each conversion, the mean and variance

associated with the ratio of the expected SBR – based on an alternative definition to the expected SBR based on the 28 weeks of gestation or more definition – was estimated. The mean was used as a bias-adjustment parameter in the model fitting, and the variance was used to account for additional uncertainty associated with the alternative definition.

Data limitations necessitated some assumptions regarding definitional adjustments. For survey data, a seven-month duration of pregnancy was assumed to be equal to a 28 weeks or more definition. Further, in LMICs, it was assumed that the SBR observed using a stillbirth definition of a birthweight of 1,000 g or more was equal to the SBR observed using the 28 weeks of gestation or more definition. Similarly, it was assumed that the SBR observed with a birthweight of 500 g or more definition equaled the SBR observed with a 22 weeks of gestational age or more definition.

► Estimation of SBRs

Estimation and projection of SBRs was undertaken using a statistical model for all country-years. In the model, the SBR was estimated assuming that the

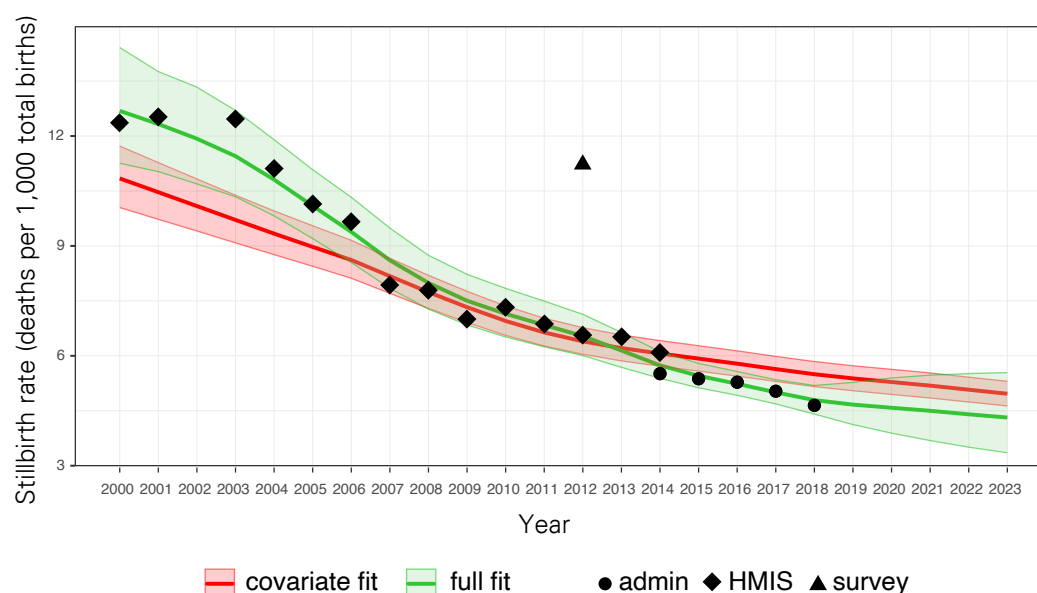
$$\text{Observed } \log(\text{SBR}) = \log(\text{true SBR}) + \text{bias} + \text{measurement error}$$

where the true SBR in a country for years 2000 to 2023 = country-intercept + SBR predicted by covariates + country-specific temporal smoothing process (explained further below). The model produced estimates of the SBR for years 2000 to 2023 with uncertainties.

True SBR component

The model for the true SBR includes three terms: 1) the country-intercept; 2) the SBR predicted by covariates; and 3) the country-specific temporal smoothing process. Covariates were used to inform SBR levels and trends, i.e., the NMR was found to be predictive of SBR, as NMR-driven estimates of the SBR were higher in country-periods with higher NMR. Figure 18 illustrates how the trend estimates (blue) are a weighted combination of information from country data and covariates. If data are precise, the SBR estimates follow the country data. In the case of no data or imprecise data, the estimates are covariate based.

Figure 18: Covariates and country data



Note: The above figure shows estimated SBR trends with 90 per cent uncertainty intervals and source data. The diamonds represent observed SBR data in the country. The red line shows the estimated SBR trend based on model covariates alone, with the uncertainty interval shown with the pink shaded area. The green line, with the uncertainty interval shown with the light green shaded area, shows the estimated SBR trend based on the country-specific data via a country-specific intercept and applying a temporal smoothing process to the red line. Note that the green line more closely fits the observed data, as it is a weighted combination of the covariate estimates and country data.

Country-year estimates can deviate from covariate-based ones through the country intercept and the temporal smoothing process. The country intercept was estimated using a multilevel model so that information on the level of the SBR is exchanged across countries within the same region.²⁷ For countries with data meeting inclusion criteria, the intercept is a weighted average of country data and the regional intercept, with weights taking account of the quantity and uncertainty associated with the country data and the variability of the estimated country intercepts. The process results in data-driven intercepts in countries with precise data. For countries without data included, the intercept is equal to the regional intercept. The temporal smoother allows deviations away from 'covariate + intercept-based' estimates based on the data so that estimates can follow precise data where available.

Bias component

The *bias* refers to the *definitional adjustment bias* + *source type bias*, where definitional adjustment bias is equal to zero for observations based on the 28 weeks of gestation or more definition and given by estimated adjustments (see section On definitional adjustment of

stillbirth data), and the source type bias is equal to zero for all observations except for observations from surveys. In the model fitting, bias terms were included to account for the bias associated with the use of definitions other than the 28 weeks of gestation or more definition and with the use of different types of data sources.

Measurement error component

To account for measurement error, varying levels of uncertainty (error variance) affect the weighting of individual observations in the model. Observations with lower error variance carry a higher weight in determining estimates as compared to observations with higher error variance.

The measurement error refers to the stochastic/sampling error + random definitional adjustment error + source type error, where each error is expected to be zero on average but has a variance term associated with it that reflects how much uncertainty is associated with the error. The stochastic/sampling error is due to observing a finite number of events and/or survey sampling design, the random definitional adjustment error which is equal to zero for observations based on the 28 weeks or more of gestation definition and non-zero otherwise. The source

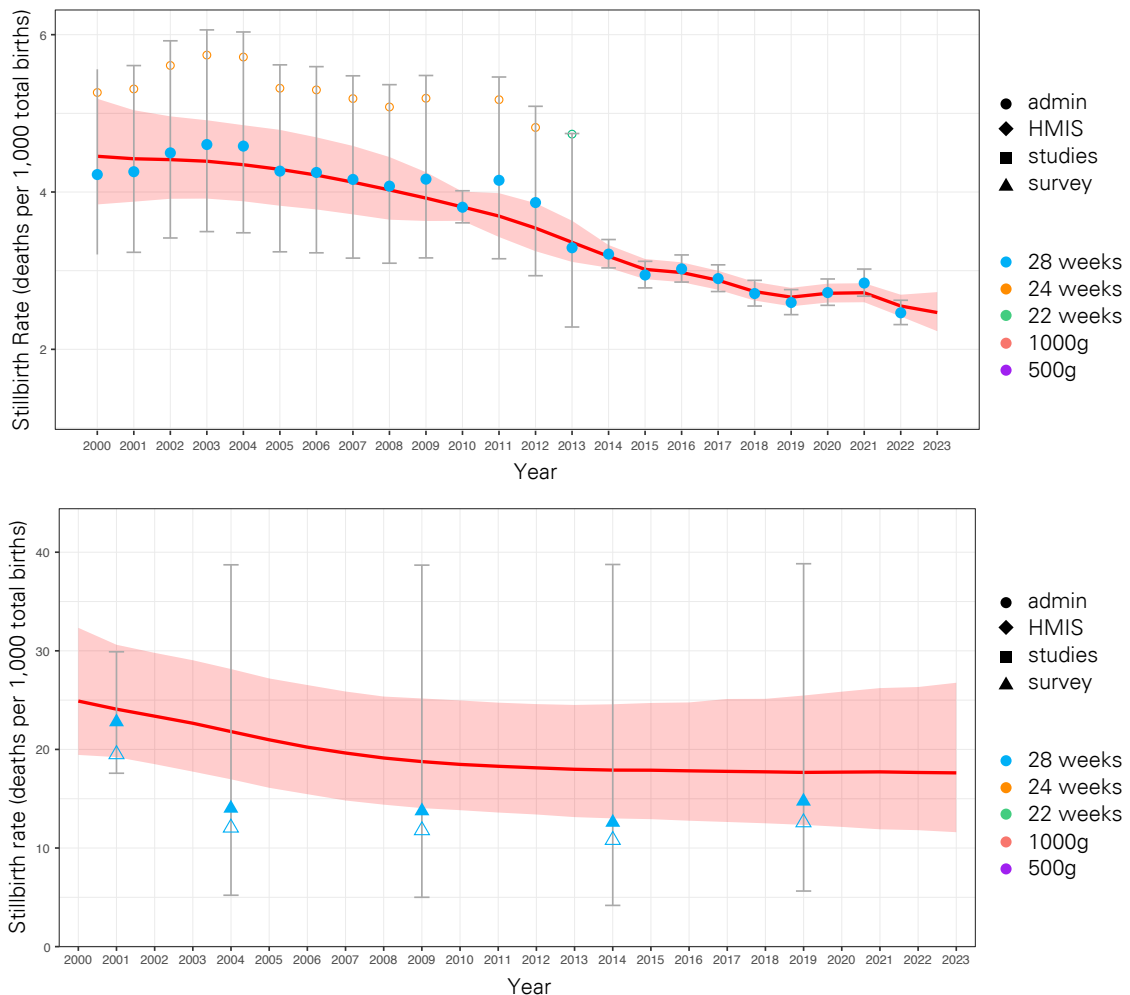
type error refers to a random error with source-type specific variance, to account for random errors that may have occurred in the data collection process and potential non-representativeness of the observation. The different data source types are 1) administrative registration, 2) HMIS, 3) household surveys, and 4) population studies.

The uncertainty associated with the measurement error in the SBR estimates depends on data availability

and precision for the respective country-period; uncertainty decreases as data availability and precision increase. Uncertainty in SBR estimates increases when extrapolating to periods without data.

Figure 19 shows the effect of varying levels of uncertainty associated with different observations. The dots show country data by definition and source type, and the vertical line illustrates the uncertainty

Figure 19: Estimated uncertainty in country data and trend estimates



Note: The above figures illustrate the bias component and measurement error incorporated into SBR estimates. The top panel shows data with definitional adjustments applied. The hollow orange and green circles show unadjusted SBR country data using a non- '28-weeks or more of gestation' definition (e.g., a 22- or 24-week definition). To use the observed, non-28-weeks SBR data in the model fitting procedure, a definitional adjustment is applied, resulting in an adjusted SBR with a 28-weeks definition. Bias-adjusted SBR data are indicated by the solid blue circles with the respective standard errors shown with the vertical bands. Note that the standard error around adjusted data points is larger than for data points where no definitional adjustment is applied (e.g., and adjusted 2012 data point versus an unadjusted 2014 data point) due to the added measurement error from bias adjustment.

The bottom panel shows data with source type adjustments applied. Household surveys have been shown to underreport SBRs; thus, observed SBRs and corresponding standard errors obtained from surveys are adjusted. The hollow triangles show the observed SBR from the survey, and the filled triangles show the adjusted SBR. The adjusted standard errors include the source type-specific measurement error and are represented with the vertical lines extending from the solid triangles.

associated with each observation. The black line is the trend estimate and the grey area represents the uncertainty. Varying levels of uncertainty (error variance) affect the uncertainty in final estimates. Observations with lower-error variance carry higher weight in determining estimates compared to observations with higher-error variance.

► Extrapolation to target year

In countries where recent stillbirth data were not available, stillbirth estimates were extrapolated from the most recent year of available national stillbirth data to 2023. Among the countries with included stillbirth data, on average 3.3 years needed to be extrapolated, with nearly half (45 per cent) of the countries having their latest data point before 2020. For 65 per cent of the countries with included data, the most recent data point referred to the last five years.

► Calculation of stillbirths

The number of stillbirths in each country is calculated using the following formula: Number of stillbirths = live births * SBR/(1-SBR). The annual estimate of the number of live births in each country from the World Population Prospects: The 2024 revision²⁸ are used along with the UN IGME SBR estimates to calculate the estimated numbers of stillbirths.

Estimation of the proportion of stillbirths that are intrapartum

Background

The intrapartum SBR is a useful marker of the quality of intrapartum (childbirth) care. An intrapartum stillbirth is a death that occurs after the onset of labour but before birth. The presence of a fetal heart rate at the onset of labour must be confirmed to diagnose intrapartum stillbirth. In settings where fetal heart rate monitoring is unavailable, assessment of the skin appearance is frequently used as a proxy to estimate stillbirth timing. As signs of skin maceration begin 6 to 12 hours after fetal death, fresh appearance of the skin with no signs of maceration is judged as a surrogate measure for intrapartum stillbirth.²⁹ However, this assessment might be unreliable and

can underestimate intrapartum stillbirth, especially when fetal death during labour occurs at home or if delays in access to care are more than 6 to 12 hours long.³⁰

► Summary of estimation approach

The UN IGME compiled all available country-level stillbirth data from administrative sources, household surveys and population-based studies to estimate country-specific trends of the proportion of stillbirths that occur in the intrapartum period. It used a smoothing time series model, adjusting for the NMR as a covariate and accounting for different definitional issues, data source types and coverage of the data available. This process averaged empirical data on the stillbirth intrapartum proportion derived from the different sources for a given country. In the case of countries with sparse or no data, regional effects and the NMR informed the level and trend in the intrapartum proportion. The estimated proportion was converted into the intrapartum SBR by multiplying the overall SBR for a particular country-year as estimated by the UN IGME. For a more detailed technical description of the estimation method, see Chong and Alexander.³¹

► Data sources

All available intrapartum subnational or national stillbirth data were derived from administrative sources or facility- or population-based research studies. Data were assessed using specified inclusion/exclusion criteria. Data points using any threshold ≥ 20 weeks or ≥ 400 g for defining stillbirth and classifying time of death using clinical assessment, skin appearance or verbal autopsy were considered for inclusion. Data points from national routine data sources (e.g., CRVS, HMIS/DHIS2) were included if the data source captured more than 80 per cent of the World Population Prospects: The 2024 revision estimated live births in the given year. Population-based studies were included if details of the birth outcome were available for 80 per cent or more of births in the population. Studies from health facilities were included if they reported birth outcomes for 80 per cent or more of births in a geographical population. In total, about 1,500 data points were collected from 92 countries. To inform intrapartum SBRs in the case of countries with little or no data, the estimation model included the NMR as a covariate in the estimation model.

► Estimation methods

For countries with data available on the proportion of stillbirths that are intrapartum, there is large variation in the quality, coverage and definitions behind these data. To reconcile these differences, the proportion of stillbirths that is intrapartum is estimated using a statistical model that accounts for various effects and adjustments, as well as estimating a smooth trend based on the available observations. The method also accounts for the varying types of uncertainty, based on the source of data, and the under-coverage of the number of stillbirths observed in the data. The method is described in more detail below.

The proportion of stillbirths that is intrapartum (IPSB) is defined as:

$$IPSB = (\text{number of stillbirths that occur in intrapartum period}) / (\text{total number of stillbirths})$$

The proportion of stillbirths that are intrapartum for a particular population in a particular year within a particular country is estimated as:

$$\text{Observed logit}(IPSB) = \text{true logit}(IPSB) + \text{bias} + \text{measurement error}$$

Where the model for the true proportion of intrapartum stillbirths contains a global-, regional-, country- and population-level intercept, the country-year-specific NMR (on the log scale) and a population-specific temporal smoothing component. The intercepts account for average levels of the proportion at various geographic hierarchies. Note that if there is only one data source for a country (e.g., a single national-level CRVS), there is no separate population effect estimated but additional uncertainty may still be added from the unobserved component described below. The NMR covariate estimates a global relationship between the IPSB and NMR, which allows estimates of the trend of IPSB to be obtained in the absence of data. Population-year estimates can deviate from the NMR-driven trend through the temporal smoothing process, which is modelled using a penalized splines model.

The proportion of stillbirths that are intrapartum was constructed using the 28-week gestation

definition where available for stillbirth and, for stillbirths using alternative definitions, the estimation model includes an adjustment factor that accounts for definitional issues in the data. A different adjustment factor is estimated based on whether a country is in a low- or high-income region. To account for varying levels of uncertainty around data observations that come from different collection systems, different error terms were estimated based on the data source. In particular, the source type error was assumed to be zero for CRVS data observations and then estimated separately for data originating from health facilities, DHIS/HMIS, and population studies. For some countries, data that exist on IPSB may only pertain to a specific subnational context, such as a certain health facility or subnational region.

In addition, observations at subnational or national levels may not capture the full extent of stillbirths occurring in that country. As such, the estimation of IPSB for each country-year involved a weighting procedure to account for the relative share of stillbirths observed from each data source as well as the share of unobserved stillbirths. In particular, the final estimate for a particular country-year is the weighted average of estimates from each population in that country, plus an unobserved component. The weights for each population were calculated based on the proportion of total stillbirths that were observed in the population. The model was fit in a Bayesian framework. Best estimates of the proportion of stillbirths that are intrapartum were taken to be the median of the posterior samples, and 90 per cent uncertainty intervals were calculated based on the 5th and 95th quantiles of the posterior samples.

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